

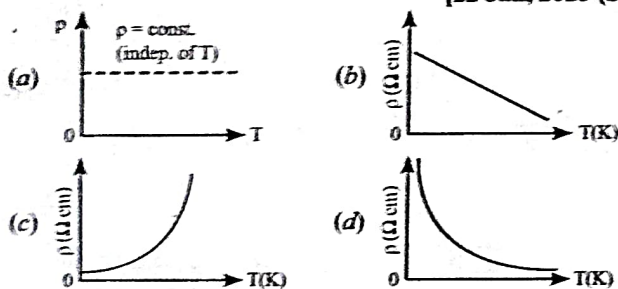
Current Electricity

Current Density, Resistance, Coefficient of Resistance, Drift Velocity, Ohm's Law and Conductivity, Resistivity

1. Current passing through a wire as function of time is given as $I(t) = 0.02t + 0.01$ A. The charge that will flow through the wire from $t = 1$ s to $t = 2$ s is: [04 April, 2025 (Shift-I)]

(a) 0.06 C (b) 0.02 C
(c) 0.07 C (d) 0.04 C

2. Which of the following resistivity (ρ) v/s temperature (T) curves is most suitable to be used in wire bound standard resistors? [22 Jan, 2025 (Shift-I)]



3. A wire of resistance 9Ω is bent to form an equilateral triangle. Then the equivalent resistance across any two vertices will be _____ ohm. [24 Jan, 2025 (Shift-I)]

(a) $27/32$ (b) $32/27$ (c) $9/8$ (d) $8/9$

4. A wire of resistance R is bent into an equilateral triangle and an identical wire is bent into a square. The ratio of resistance between the two end points of an edge of the triangle to that of the square is [28 Jan, 2025 (Shift-I)]

(a) $27/32$ (b) $32/27$ (c) 800 C (d) 1600 C

5. The electric current through a wire varies with time as $I = I_0 + \beta t$, where $I_0 = 20$ A and $\beta = 3$ A/s. The amount of electric charge crossed through a section of the wire in 20 s is: [29 Jan, 2024 (Shift-I)]

(a) 694°C (b) 1235°C (c) 1694°C (d) 1667°C

6. An electric toaster has resistance of 60Ω at room temperature (27°C). The toaster is connected to a 220 V supply. If the current flowing through it reaches 2.75 A, the temperature attained by toaster is around: (if $\alpha = 2 \times 10^{-4}/^\circ\text{C}$) [30 Jan, 2024 (Shift-I)]

are: [31 Jan, 2024 (Shift-I)]

(a) $\alpha_1 + \alpha_2, \frac{\alpha_1 + \alpha_2}{2}$ (b) $\frac{\alpha_1 + \alpha_2}{2}, \frac{\alpha_1 + \alpha_2}{2}$

(c) $\alpha_1 + \alpha_2, \frac{\alpha_1 \alpha_2}{\alpha_1 + \alpha_2}$ (d) $\frac{\alpha_1 + \alpha_2}{2}, \alpha_1 + \alpha_2$

8. The current in a conductor is expressed as $I = 3t^2 + 4t^3$, where I is in Ampere and t is in second. The amount of electric charge that flows through a section of the conductor during $t = 1$ s to $t = 2$ s is _____ C. [1 Feb, 2024 (Shift-I)]

9. The resistances of the platinum wire of a platinum resistance thermometer at the ice point and steam point are 8Ω and 10Ω respectively. After inserting in a hot bath of temperature 400°C , the resistance of platinum wire is: [04 April, 2024 (Shift-I)]

(a) 2Ω (b) 16Ω (c) 8Ω (d) 10Ω

10. Two wires A and B are made up of the same material and have the same mass. Wire A has radius of 2.0 mm and wire B has radius of 4.0 mm. The resistance of wire B is 2Ω . The resistance of wire A is _____ Ω . [04 April, 2024 (Shift-II)]

11. A wire of resistance R and radius r is stretched till its radius became $r/2$. If new resistance of the stretched wire is xR , then value of x is _____. [06 April, 2024 (Shift-I)]

12. Resistance of a wire at 0°C , 100°C and $t^\circ\text{C}$ is found to be 10Ω , 10.2Ω and 10.95Ω respectively. The temperature t in Kelvin scale is _____. [08 April, 2024 (Shift-I)]

13. At room temperature (27°C), the resistance of a heating element is 50Ω . The temperature coefficient of the material is $2.4 \times 10^{-4}/^\circ\text{C}$. The temperature of the element, when its resistance is 62Ω , is _____. [09 April, 2024 (Shift-II)]

14. The length of a given cylindrical wire is increased to double of its original length. The percentage increase in the resistance of the wire will be _____. [25 June, 2022 (Shift-I)]

15. An aluminium wire is stretched to make its length, 0.4% larger. The percentage change in resistance is: [26 June, 2022 (Shift-II)]

(a) 0.4% (b) 0.2%
(c) 0.8% (d) 0.6%

16. The current density in a cylindrical wire of radius 4 mm is 4×10^6 Am^{-2} . The current through the outer portion of the wire between radial distance $\frac{R}{2}$ and R is _____ πA . [27 June, 2022 (Shift-I)]

17. The current density in a cylindrical wire of radius $r = 4.0 \text{ mm}$ is $1.0 \times 10^6 \text{ A/m}^2$. The current through the outer portion of the wire between radial distances $\frac{r}{2}$ and r is $x\pi A$; where x is _____

[27 June, 2022 (Shift-II)]

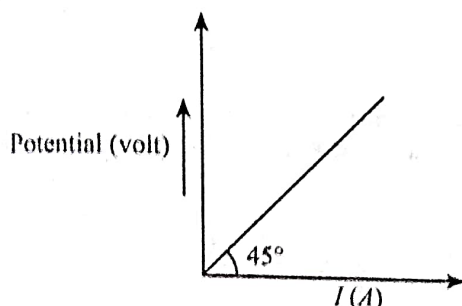
18. Resistance of the wire is measured as 2Ω and 3Ω at 10°C and 30°C respectively. Temperature co-efficient of resistance of the material of the wire is:

[28 June, 2022 (Shift-II)]

19. The variation of applied potential and current flowing through a given wire is shown in figure. The length of wire is 31.4 cm . The diameter of wire is measured as 2.4 cm . The resistivity of the given wire is measured as $x \times 10^{-3} \Omega\text{m}$. The value of x is _____

[29 June, 2022 (Shift-I)]

[Take $\pi = 3.14$]



20. A wire of resistance R_1 is drawn out so that its length is increased by twice of its original length. The ratio of new resistance to original resistance is

[28 July, 2022 (Shift-I)]

- (a) 9 : 1 (b) 1 : 9
(c) 4 : 1 (d) 3 : 1

21. Given below are two statements: one is labelled as Assertion A and the other is labelled as Reason R.

Assertion (A): Alloys such as constantan and manganin are used in making standard resistance coils.

Reason (R): Constantan and manganin have very small value of temperature coefficient of resistance.

In the light of the above statements, choose the correct answer from the options given below.

[29 July, 2022 (Shift-II)]

- (a) Both (A) and (R) are true and (R) is the correct explanation of A
(b) Both (A) and (R) are true but (R) is NOT the correct explanation of A
(c) (A) is true but (R) is false
(d) (A) is false but (R) is true

22. A 1 m long wire is broken into two unequal parts X and Y . The X part of the wire is stretched into another wire W . Length of W is twice the length of X and the resistance of W is twice that of Y . Find the ratio of length of X and Y .

[29 July, 2022 (Shift-II)]

- (a) 1 : 4 (b) 1 : 2
(c) 4 : 1 (d) 2 : 1

23. A current through a wire depends on time as $i = \alpha_0 t + \beta t^2$ Where $\alpha_0 \approx 20 \text{ A/s}$ and $\beta = 8 \text{ As}^{-2}$. Find the charge crossed through a section of the wire in 15 s .

[24 Feb, 2021 (Shift-I)]

- (a) 2100 C (b) 2250 C (c) 260 C (d) 11250 C

24. A cylindrical wire of radius 0.5 mm and conductivity $5 \times 10^7 \text{ S/m}$ is subjected to an electric field of 10 mV/m . The expected value of current in the wire will be $x^3 \pi \text{ mA}$. The value of x is _____

[24 Feb, 2021 (Shift-II)]

25. A current of 10 A exists in a wire of cross-sectional area of 5 mm^2 with a drift velocity of $2 \times 10^{-3} \text{ ms}^{-1}$. The number of free electrons in each cubic meter of the wire

[17 March, 2021 (Shift-I)]

- (a) 2×10^{25} (b) 1×10^{23}
(c) 2×10^6 (d) 625×10^{23}

26. A current of 5 A is passing through a non-linear magnesium wire of cross-section 0.04 m^2 . At every point the direction of current density is at an angle of 60° with the unit vector of area of cross-section. The magnitude of electric field at every point of the conductor is (Resistivity of magnesium, $\rho = 44 \times 10^{-8} \Omega\text{m}$)

[20 July, 2021 (Shift-I)]

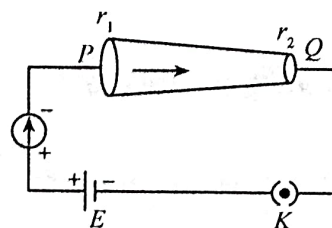
- (a) $11 \times 10^{-7} \text{ V/m}$ (b) $11 \times 10^{-2} \text{ V/m}$
(c) $11 \times 10^{-5} \text{ V/m}$ (d) $11 \times 10^{-3} \text{ V/m}$

27. In the given figure, a battery of emf E is connected across a conductor PQ of length ' l ' and different area of cross-sections by having radii r_1 and r_2 ($r_2 < r_1$).

Choose the correct option as one moves from P to Q :

[27 July, 2021 (Shift-I)]

- (a) All of these.
(b) Electron current decreases.
(c) Drift velocity of electron increases.
(d) Electric field decreases.



28. The resistance of a conductor at 15°C is 16Ω and at 100°C is 20Ω . What will be the temperature coefficient of resistance of the conductor?

[27 July, 2021 (Shift-II)]

- (a) 0.003°C^{-1} (b) 0.042°C^{-1} (c) 0.033°C^{-1} (d) 0.010°C^{-1}

29. Consider four conducting materials copper, tungsten, mercury and aluminium with resistivity ρ_C , ρ_T , ρ_M and ρ_A respectively. Then

[2 Sep, 2020 (Shift-I)]

- (a) $\rho_A > \rho_T > \rho_C$ (b) $\rho_M > \rho_A > \rho_C$
(c) $\rho_C > \rho_A > \rho_T$ (d) $\rho_A > \rho_M > \rho_C$

30. A circuit to verify Ohm's law uses ammeter and voltmeter in series or parallel connected correctly to the resistor. In the circuit:

[2 Sep, 2020 (Shift-I)]

- (a) ammeter is always used in parallel and voltmeter is series.
(b) both ammeter and voltmeter must be connected in parallel.
(c) ammeter is always connected in series and voltmeter in parallel.
(d) both, ammeter and voltmeter must be connected in series.

31. Drift speed of electrons, when 1.5 A of current flows in a copper wire of cross section 5 mm^2 , is v . If the electron density in copper is $9 \times 10^{28} \text{ m}^{-3}$ the value of v in mm/s is close to (Take charge of electron to be $= 1.6 \times 10^{-19} \text{ C}$)

[9 Jan, 2019 (Shift-I)]

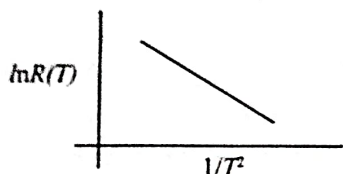
- (a) 0.02 (b) 3 (c) 2 (d) 0.2

32. In a conductor, if the number of conduction electrons per unit volume is $8.5 \times 10^{28} \text{ m}^{-3}$ and mean free time is 25 fs (femto second), its approximate resistivity is: ($m_e = 9.1 \times 10^{-31} \text{ kg}$)

[9 Apr, 2019 (Shift-II)]

- (a) $10^{-5} \Omega\text{m}$ (b) $10^{-6} \Omega\text{m}$ (c) $10^{-7} \Omega\text{m}$ (d) $10^{-8} \Omega\text{m}$

33. In an experiment, the resistance of a material is plotted as a function of temperature (in some range). As shown in the figure, it is a straight line. One may conclude that: [10 April, 2019 (Shift-I)]



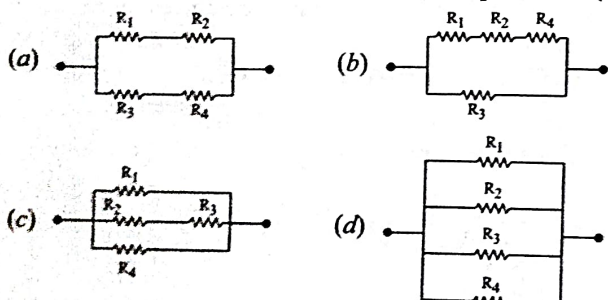
- (a) $R(T) = R_0 e^{T^{-1}/T_0^{-1}}$ (b) $R(T) = \frac{R_0}{T^2}$
 (c) $R(T) = R_0 e^{-T^{-1}/T_0^{-1}}$ (d) $R(T) = R_0 e^{-(T_0^2/T^2)}$
34. A current of 5 A passes through a copper conductor (resistivity $= 1.7 \times 10^{-8} \Omega\text{m}$) of radius of cross-section 5 mm. Find the mobility of the charges if their drift velocity is $1.1 \times 10^{-3} \text{ m/s}$. [10 April, 2019 (Shift-I)]

- (a) $1.8 \text{ m}^2/\text{Vs}$ (b) $1.0 \text{ m}^2/\text{Vs}$ (c) $1.3 \text{ m}^2/\text{Vs}$ (d) $1.5 \text{ m}^2/\text{Vs}$
35. Space between two concentric conducting spheres of radii a and b ($b > a$) is filled with a medium of resistivity ρ . The resistance between the two spheres will be [10 April, 2019 (Shift-II)]

- (a) $\frac{\rho}{2\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$ (b) $\frac{\rho}{2\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$
 (c) $\frac{\rho}{4\pi} \left(\frac{1}{a} + \frac{1}{b} \right)$ (d) $\frac{\rho}{4\pi} \left(\frac{1}{a} - \frac{1}{b} \right)$

Series and Parallel Combination of Resistance

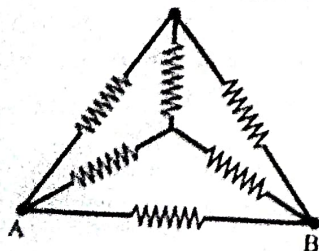
36. From the combination of resistors with resistance values $R_1 = R_2 = R_3 = 5\Omega$ and $R_4 = 10\Omega$, which of the following combination is the best circuit to get an equivalent resistance of 6Ω ? [04 April, 2025 (Shift-II)]



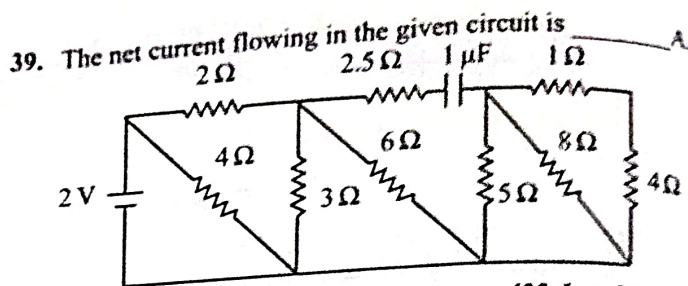
37. A wire of length 25 m and cross-sectional area 5 mm^2 having resistivity of $2 \times 10^{-6} \Omega$ is bent into a complete circle. The resistance between diametrically opposite points will be [03 April, 2025 (Shift-I)]

- (a) 12.5Ω (b) 50Ω (c) 100Ω (d) 25Ω

38. A wire of resistance R is bent into a triangular pyramid as shown in figure with each segment having same length. The resistance between points A and B is R/n . The value of n is: [07 April, 2025 (Shift-I)]

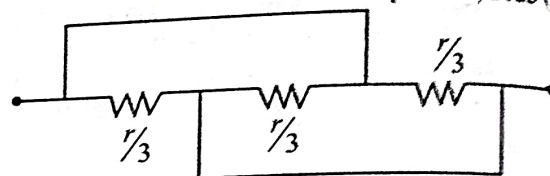


- (a) 16 (b) 14 (c) 10 (d) 12



[22 Jan, 2025 (Shift-I)]

40. Find the equivalent resistance between two ends of the following circuit [28 Jan, 2025 (Shift-I)]

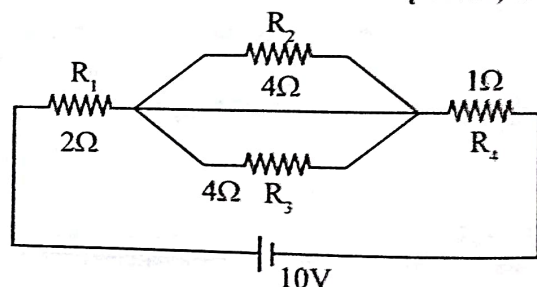


- (a) $\frac{r}{6}$ (b) $\frac{r}{9}$ (c) r (d) $\frac{r}{3}$

41. A wire of resistance R and length L is cut into 5 equal parts. If these parts are joined parallelly, then resultant resistance will be: [27 Jan, 2024 (Shift-I)]

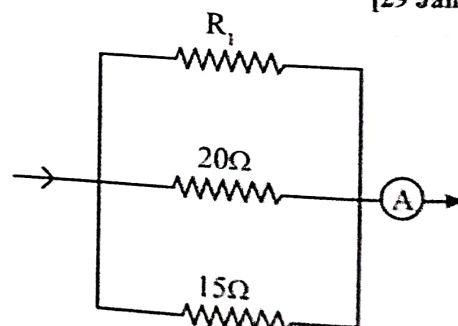
- (a) $\frac{1}{25} R$ (b) $\frac{1}{5} R$ (c) $25 R$ (d) $5 R$

42. In the given circuit, the current in resistance R_3 is: [29 Jan, 2024 (Shift-I)]

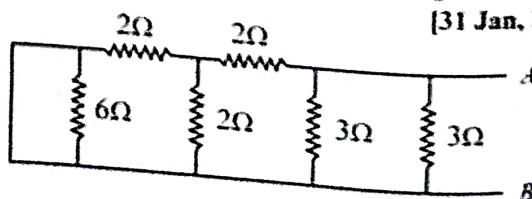


- (a) 1 A (b) 1.5 A (c) 2 A (d) 2.5 A

43. In the given circuit, the current flowing through the resistance R_1 is 0.3 A, while the ammeter reads 0.9 A. The value of R_1 is Ω . [29 Jan, 2024 (Shift-I)]



44. Equivalent resistance of the following network is Ω [31 Jan, 2024 (Shift-I)]

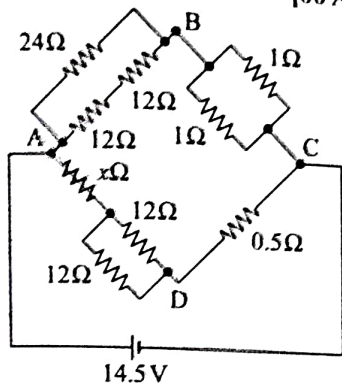


46. A wire of resistance $20\ \Omega$ is divided into 10 equal parts. A combination of two parts are connected in parallel and so on. Now resulting pairs of parallel combination are connected in series. The equivalent resistance of final combination is $\underline{\hspace{2cm}}\ \Omega$.

[05 April, 2024 (Shift-II)]

47. The value of unknown resistance (x) for which the potential difference between B and D will be zero in the arrangement shown, is:

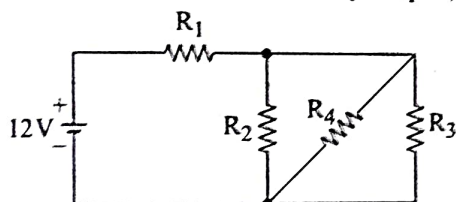
[06 April, 2024 (Shift-I)]



- (a) $3\ \Omega$ (b) $9\ \Omega$
(c) $6\ \Omega$ (d) $42\ \Omega$

47. In the given figure $R_1 = 10\ \Omega$, $R_2 = 8\ \Omega$, $R_3 = 4\ \Omega$ and $R_4 = 8\ \Omega$. Battery is ideal with emf 12V . Equivalent resistance of the circuit and current supplied by battery are respectively.

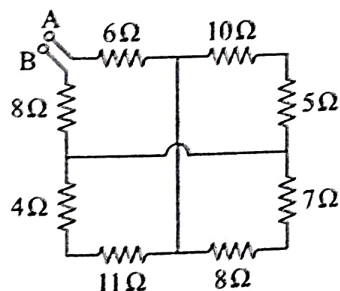
[05 April, 2024 (Shift-I)]



- (a) $12\ \Omega$ and $11.4\ \text{A}$ (b) $10.5\ \Omega$ and $1.14\ \text{A}$
(c) $10.5\ \Omega$ and $1\ \text{A}$ (d) $12\ \Omega$ and $1\ \text{A}$

48. The equivalent resistance between A and B is:

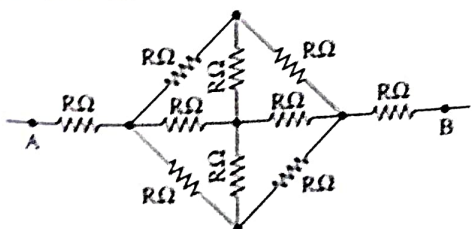
[09 April, 2024 (Shift-I)]



- (a) $18\ \Omega$ (b) $25\ \Omega$
(c) $27\ \Omega$ (d) $19\ \Omega$

49. The effective resistance between A and B , if resistance of each resistor is R , will be

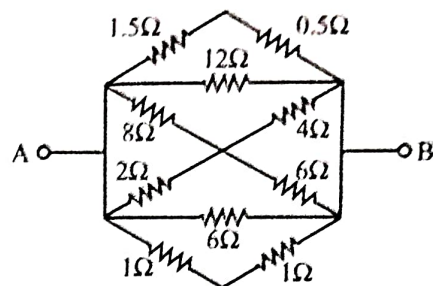
[09 April, 2024 (Shift-II)]



- (a) $\frac{2}{3}R$ (b) $\frac{8R}{3}$ (c) $\frac{5R}{3}$ (d) $\frac{4R}{3}$

50. The equivalent resistance between A and B is $\underline{\hspace{2cm}}$

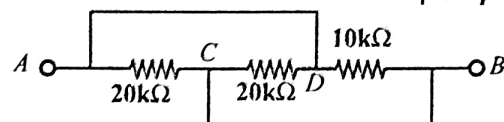
[30 Jan, 2023 (Shift-II)]



- (a) $\frac{2}{3}\ \Omega$ (b) $\frac{1}{2}\ \Omega$ (c) $\frac{3}{2}\ \Omega$ (d) $\frac{1}{3}\ \Omega$

51. The equivalent resistance between A and B as shown in figure is:

[08 April, 2023 (Shift-II)]



- (a) $5\ \text{k}\Omega$ (b) $30\ \text{k}\Omega$ (c) $10\ \text{k}\Omega$ (d) $20\ \text{k}\Omega$

52. Equivalent resistance between the adjacent corners of a regular n -sided polygon of uniform wire of resistance R would be:

[1 Feb, 2023 (Shift-II)]

- (a) $\frac{(n-1)R}{n^2}$ (b) $\frac{(n-1)R}{(2n-1)}$ (c) $\frac{n^2R}{n-1}$ (d) $\frac{(n-1)R}{n}$

53. Given below are two statements:

Statement-I: The equivalent resistance of resistors in a series combination is smaller than least resistance used in the combination.

Statement-II: The resistivity of the material is independent of temperature.

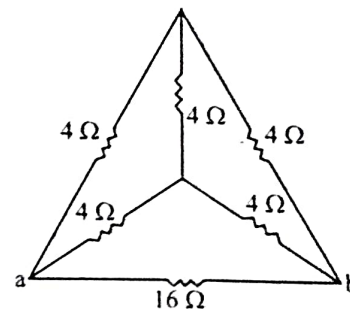
In the light of the above statements, choose the correct answer from the options given below:

[15 April, 2023 (Shift-I)]

- (a) Statement-I is false but Statement-II is true
(b) Both Statement-I and Statement-II are false
(c) Statement-I is true but Statement-II is false
(d) Both Statement-I and Statement-II are true

54. The equivalent resistance of the circuit shown below between points a and b is:

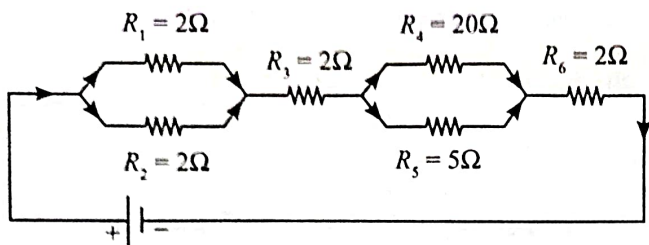
[10 April, 2023 (Shift-I)]



- (a) $24\ \Omega$ (b) $3.2\ \Omega$ (c) $20\ \Omega$ (d) $16\ \Omega$

55. As shown in the figure, a network of resistors is connected to a battery of 24V with an internal resistance of $3\ \Omega$. The currents through the resistors R_1 and R_2 are I_1 and I_2 respectively. The values of I_1 and I_2 are:

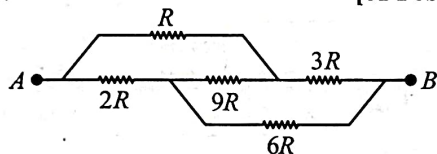
[24 Jan, 2023 (Shift-I)]



(24V, 3Ω)

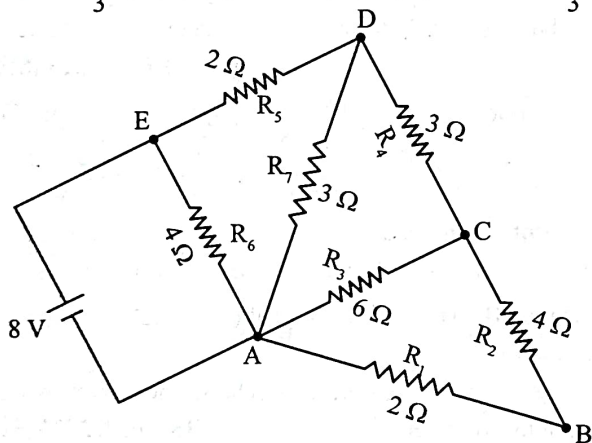
- (a) $I_4 = \frac{8}{5} A$ and $I_5 = \frac{2}{5} A$ (b) $I_4 = \frac{24}{5} A$ and $I_5 = \frac{6}{5} A$
 (c) $I_4 = \frac{6}{5} A$ and $I_5 = \frac{24}{5} A$ (d) $I_4 = \frac{2}{5} A$ and $I_5 = \frac{8}{5} A$

56. The equivalent resistance between A and B of the network shown in figure: [01 Feb, 2023 (Shift-I)]



- (a) $11\frac{2R}{3}$ (b) $14R$ (c) $21R$ (d) $\frac{8}{3}R$

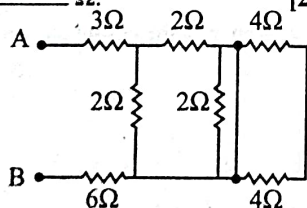
57.



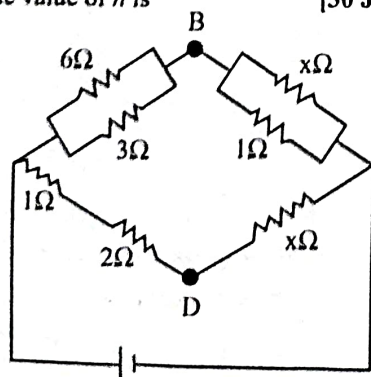
The current flowing through R_2 is: [11 April, 2023 (Shift-II)]

- (a) $\frac{2}{3} A$ (b) $\frac{1}{4} A$ (c) $\frac{1}{2} A$ (d) $\frac{1}{3} A$

58. In the given circuit, the equivalent resistance between the terminal A and B is $\frac{1}{n} \Omega$. [25 Jan, 2023 (Shift-I)]



59. If the potential difference between B and D is zero, the value of x is $\frac{1}{n} \Omega$. The value of n is [30 Jan, 2023 (Shift-II)]



60. 10 resistors each of resistance 10Ω can be connected in such as to get maximum and minimum equivalent resistance. The ratio of maximum and minimum equivalent resistance will be [10 April, 2023 (Shift-I)]

61. Two metallic wires of identical dimensions are connected in series. If σ_1 and σ_2 are the conductivities of these wires respectively, the effective conductivity of the combination is: [29 July, 2022 (Shift-I)]

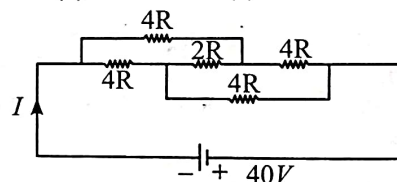
- (a) $\frac{\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (b) $\frac{2\sigma_1 \sigma_2}{\sigma_1 + \sigma_2}$ (c) $\frac{\sigma_1 + \sigma_2}{2\sigma_1 \sigma_2}$ (d) $\frac{\sigma_1 + \sigma_2}{\sigma_1 \sigma_2}$

62. Eight copper wire of length l and diameter d are joined in parallel to form a single composite conductor of resistance R . If a single copper wire of length $2l$ have the same resistance (R) then its diameter will be $\frac{1}{n}d$. [25 July, 2022 (Shift-I)]

63. What will be the most suitable combination of three resistors $A = 2\Omega$, $B = 4\Omega$, $C = 6\Omega$ so that $\left(\frac{22}{3}\right)$ is equivalent resistance of combination? [24 June, 2022 (Shift-II)]

- (a) Parallel combination of A and C connected in series with B.
 (b) Parallel combination of A and B connected in series with C.
 (c) Series combination of A and C connected in parallel with B.
 (d) Series combination of B and C connected in parallel with A.

64. The current I in the given circuit will be: [26 July, 2022 (Shift-I)]
 (a) 10 A (b) 20 A (c) 4 A (d) 40 A



65. Given below are two statements:

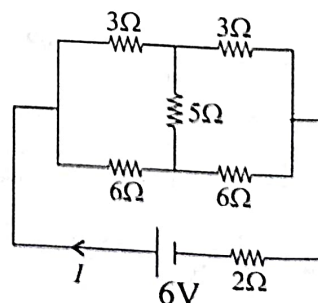
Statement-I: A uniform wire of resistance 80Ω is cut into four equal parts. These parts are now connected in parallel. The equivalent resistance of the combination will be 5Ω .

Statement-II: Two resistances $2R$ and $3R$ are connected in parallel in a electric circuit. The value of thermal energy developed in $3R$ and $2R$ will be in the ratio 3 : 2.

In the light of the above statements, choose the most appropriate answer from the option given below [28 July, 2022 (Shift-II)]

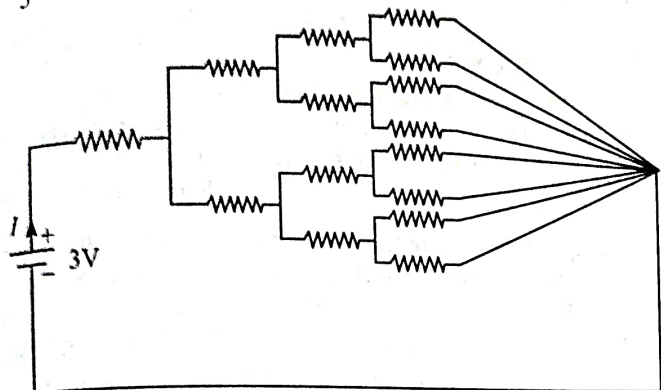
- (a) Both statement-I and statement-II are correct
 (b) Both statement-I and statement-II are incorrect
 (c) Statement-I is correct but statement-II is incorrect
 (d) Statement-I is incorrect but statement-II is correct

66. A battery of 6V is connected to the circuit as shown below. The current I drawn from the battery is: [26 July, 2022 (Shift-II)]

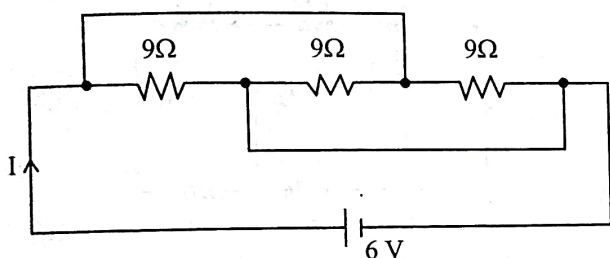


- (a) 1A (b) 2A (c) $\frac{6}{11} A$ (d) $\frac{4}{3} A$

67. All resistances in figure are $1\ \Omega$ each. The value of current 'I' is $\frac{a}{5}\text{ A}$. The value of a is _____. [28 June, 2022 (Shift-II)]

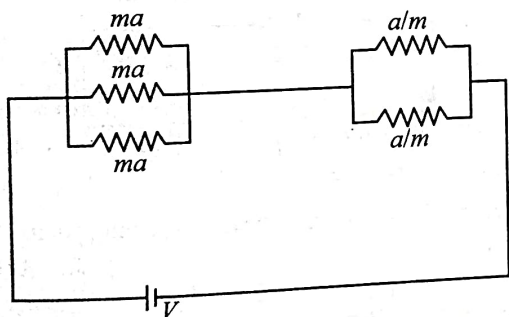


68. The current I flowing through the given circuit will be _____ A. [29 July, 2022 (Shift-I)]



69. In the given circuit 'a' is an arbitrary constant. The value of m for which the equivalent circuit resistance is minimum, will be $\sqrt{\frac{x}{2}}$.

The value of x is _____ [27 June, 2022 (Shift-II)]

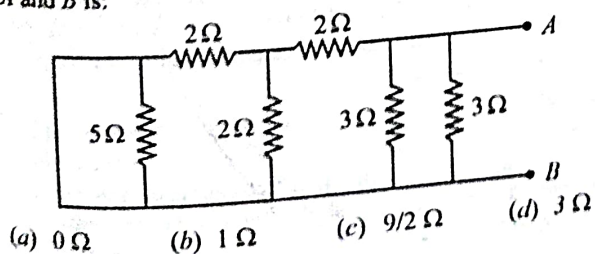


70. If you are provided a set of resistances $2\ \Omega$, $4\ \Omega$, $6\ \Omega$ and $8\ \Omega$. Connect these resistances so as to obtain an equivalent resistance of $\frac{46}{3}\ \Omega$.

[26 Aug, 2021 (Shift-II)]

- (a) $4\ \Omega$ and $6\ \Omega$ are in parallel with $2\ \Omega$ and $8\ \Omega$ in series.
 (b) $6\ \Omega$ and $8\ \Omega$ are in parallel with $2\ \Omega$ and $4\ \Omega$ in series.
 (c) $2\ \Omega$ and $4\ \Omega$ are in parallel with $6\ \Omega$ and $8\ \Omega$ in series.
 (d) $2\ \Omega$ and $6\ \Omega$ are in parallel with $4\ \Omega$ and $8\ \Omega$ in series.

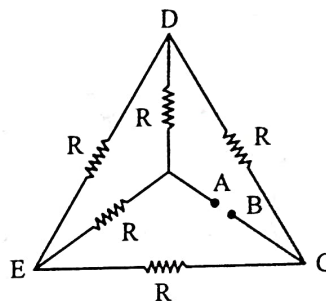
71. The equivalent resistance of the given circuit between the terminals A and B is: [31 Aug, 2021 (Shift-II)]



- (a) $0\ \Omega$ (b) $1\ \Omega$ (c) $9/2\ \Omega$ (d) $3\ \Omega$

72. Five equal resistances are connected in a network as shown in figure. The net resistance between the points A and B is: [26 Feb, 2021 (Shift-I)]

- (a) $\frac{3R}{2}$ (b) R
 (c) $\frac{R}{2}$ (d) $2R$

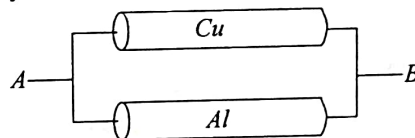


73. A Copper (Cu) rod of length 25cm and cross-sectional area 3 mm^2 is joined with a similar aluminium (Al) rod as shown in figure. Find the resistance of the combination between the ends A and B.

[22 July, 2021 (Shift-II)]

(Take Resistivity of Copper = $1.7 \times 10^{-8}\ \Omega\text{m}$,

Resistivity of Aluminium = $2.6 \times 10^{-8}\ \Omega\text{m}$)



- (a) $0.858\text{ m}\Omega$ (b) $0.0858\text{ m}\Omega$
 (c) $1.420\text{ m}\Omega$ (d) $2.170\text{ m}\Omega$

74. What equal length of an iron wire and a copper-nickel alloy wire, each of 2 mm diameter connected parallel to give an equivalent resistance of $3\ \Omega$? [26 Aug, 2021 (Shift-I)]

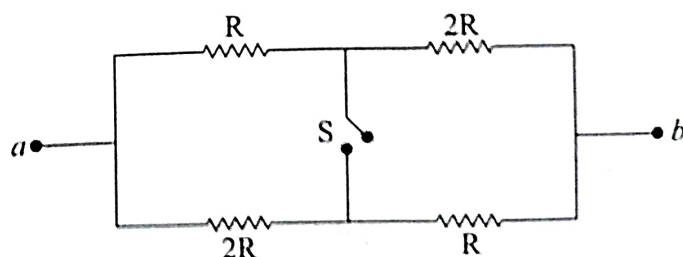
(Given resistivities of iron and copper-nickel alloy wire are $12\ \mu\Omega\text{ cm}$ and $51\ \mu\Omega\text{ cm}$ respectively)

- (a) 110 m (b) 90 m
 (c) 82 m (d) 97 m

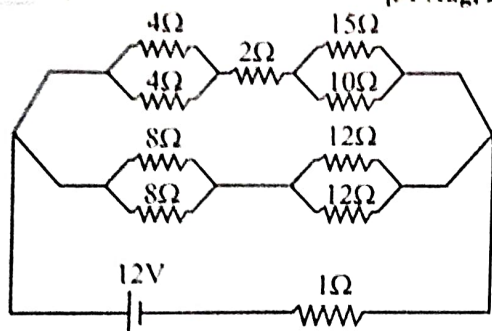
75. Two resistors $R_1 = (4 \pm 0.8)\ \Omega$ and $R_2 = (4 \pm 0.4)\ \Omega$ are connected in parallel. The equivalent resistance of their parallel combination will be: [1 Sep, 2021 (Shift-II)]

- (a) $(4 \pm 0.4)\ \Omega$ (b) $(2 \pm 0.3)\ \Omega$
 (c) $(4 \pm 0.3)\ \Omega$ (d) $(2 \pm 0.4)\ \Omega$

76. The ratio of the equivalent resistance of the network (shown in figure) between the points a and b when switch is open and switch is closed is $x : 8$. The value of x is _____. [27 Aug, 2021 (Shift-II)]

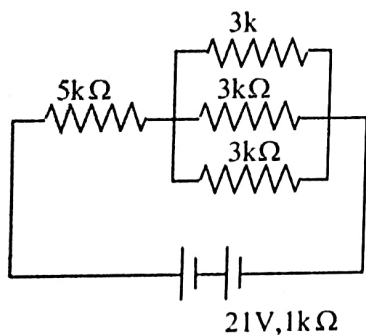


77. The voltage drop across $15\ \Omega$ resistance in the given figure will be _____ V. [31 Aug, 2021 (Shift-I)]

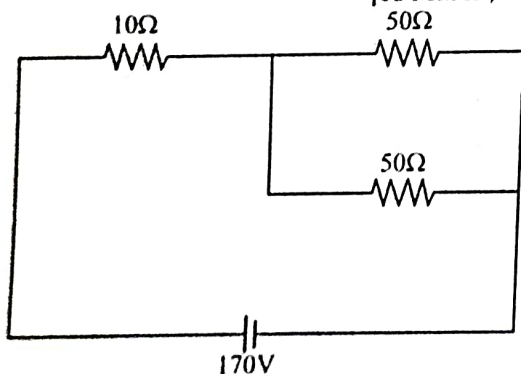


78. A square shaped wire with resistance of each side $3\ \Omega$ is bent to form a complete circle. The resistance between two diametrically opposite points of the circle in unit of Ω will be _____. [31 Aug, 2021 (Shift-I)]

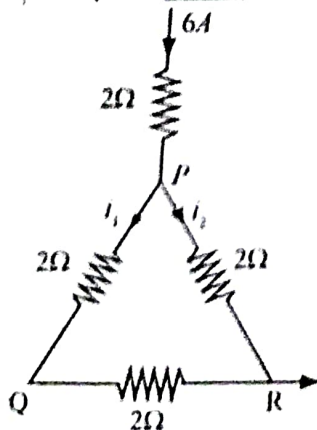
79. In the figure given, the electric current flowing through the $5\ \text{k}\Omega$ resistor is 'x' mA. The value of x to the nearest integer is _____. [16 March, 2021 (Shift-I)]



80. The voltage across the $10\ \Omega$ resistor in the given circuit is x volt. The value of 'x' to the nearest integer is _____. [18 March, 2021 (Shift-I)]



81. A current of $6\ \text{A}$ enters one corner P of an equilateral triangle PQR having 3 wires of resistance $2\ \Omega$ each and leaves by the corner R. The currents i_1 in ampere is _____. [25 Feb, 2021 (Shift-II)]

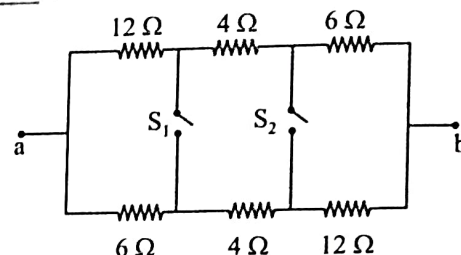


82. First, a set of n equal resistors of $10\ \Omega$ each are connected in series to a battery of emf $20\ \text{V}$ and internal resistance $10\ \Omega$. A current I is observed to flow. Then, the n resistors are connected in parallel to the same battery. It is observed that the current is increased 20 times. then the value of n is _____. [27 Aug, 2021 (Shift-I)]

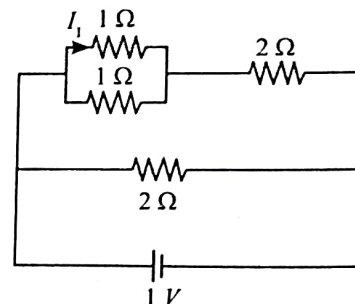
83. Two wires of same length and thickness having specific resistances $6\ \Omega\ \text{cm}$ and $3\ \Omega\ \text{cm}$ respectively are connected in parallel. The effective resistivity is $\rho\ \Omega\ \text{cm}$. The value of ρ to the nearest integer is _____. [18 March, 2021 (Shift-II)]

84. The equivalent resistance of series combination of two resistors is 's'. When they are connected in parallel, the equivalent resistance is 'p'. If $s = np$, then the minimum value for n is _____. (Round off to the nearest integer) [17 March, 2021 (Shift-I)]

85. In the given figure switches S_1 and S_2 are in open condition. The resistance across ab when the switches S_1 and S_2 are closed is _____. [20 July, 2021 (Shift-II)]

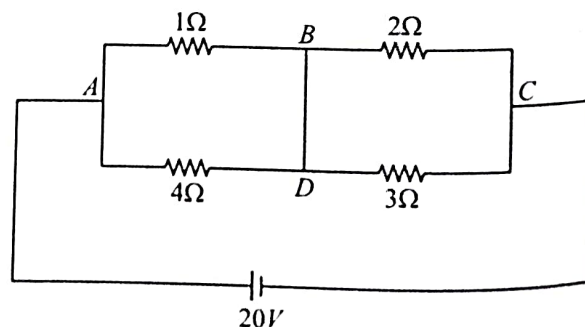


86. The current I_1 (in A) flowing through $1\ \Omega$ resistor in the following circuit is _____. [7 Jan, 2020 (Shift-I)]



- (a) 0.25 (b) 0.4 (c) 0.2 (d) 0.5

87. In the given circuit diagram, a wire is joining points B and D. The current in this wire is _____. [9 Jan, 2020 (Shift-I)]



- (a) Zero (b) 2 A (c) 0.4 A (d) 4 A

88. Four resistances of $15\ \Omega$, $12\ \Omega$, $14\ \Omega$ and $10\ \Omega$ respectively in cyclic order to form Whetstone's network. The resistance that is to be connected in parallel with the resistance of $10\ \Omega$ to balance the network is _____. [9 April, 2019 (Shift-II)]

89. A metal wire of resistance $3\ \Omega$ is elongated to make a uniform wire of double its previous length. The new wire is now bent and the ends joined to make a circle. If two points on this circle make an angle 60° at the centre, the equivalent resistance between these two points will be _____. [9 April, 2019 (Shift-II)]

- (a) $\frac{12}{5} \Omega$ (b) $\frac{5}{3} \Omega$ (c) $\frac{5}{2} \Omega$ (d) $\frac{7}{2} \Omega$

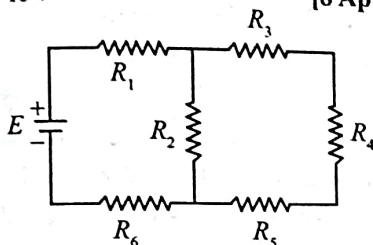
90. A uniform metallic wire has a resistance of 18Ω and is bent into an equilateral triangle. Then, the resistance between any two vertices of the triangle is:

- [10 Jan, 2019 (Shift-I)]
(a) 4Ω (b) 8Ω (c) 12Ω (d) 2Ω

91. In the figure shown, what is the current (in Ampere) drawn from the battery? You are given

- $R_1 = 15 \Omega$, $R_2 = 10 \Omega$, $R_3 = 20 \Omega$, $R_4 = 5 \Omega$, $R_5 = 25 \Omega$, $R_6 = 30 \Omega$, $E = 15 \text{ V}$

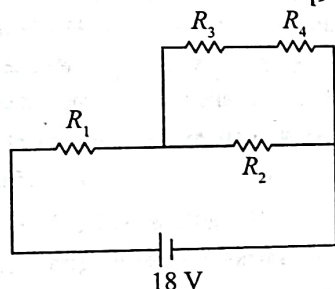
[8 April, 2019 (Shift-II)]



- (a) $13/24$ (b) $7/18$ (c) $9/32$ (d) $20/3$

92. In the given circuit the internal resistance of the 18 V cell is negligible. If $R_1 = 400 \Omega$, $R_3 = 100 \Omega$ and $R_4 = 500 \Omega$ and the reading of an ideal voltmeter across R_4 is 5 V , then the value of R_2 will be

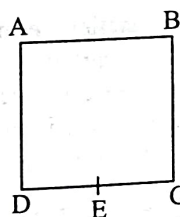
[9 Jan, 2019 (Shift-II)]



- (a) 300Ω (b) 450Ω (c) 550Ω (d) 230Ω

93. A wire of resistance R is bent to form a square ABCD as shown in the figure. The effective resistance between E and C is: (E is mid-point of arm CD)

[9 April, 2019 (Shift-I)]



- (a) $\frac{1}{16} R$ (b) $\frac{7}{64} R$ (c) $\frac{3}{4} R$ (d) R

Power and Heat Dissipated in Electric Circuits

94. There are ' n ' number of identical electric bulbs, each is designed to draw a power p independently from the mains supply. They are now joined in series across the mains supply. The total power drawn by the combination is:

[04 April, 2025 (Shift-II)]

- (a) np (b) $\frac{p}{n^2}$ (c) $\frac{p}{n}$ (d) p

95. A motor operating on 100 V draws a current of 1 A . If the efficiency of the motor is 91.6% , then the loss of power in units of cal/s is

[03 April, 2025 (Shift-II)]

- (a) 4 (b) 8.4 (c) 2 (d) 6.2

96. When a potential difference V is applied across a wire of resistance R , it dissipates energy at a rate W . If the wire is cut into two halves and these halves are connected mutually parallel across the same supply, the energy dissipation rate will become:

[30 Jan, 2024 (Shift-II)]

- (a) $1/4 W$ (b) $1/2 W$ (c) $2 W$ (d) $4 W$

97. By what percentage will the illumination of the lamp decrease if the current drops by 20% ?

[31 Jan, 2024 (Shift-II)]

- (a) 46% (b) 26% (c) 36% (d) 56%

98. In the following circuit, the battery has an emf of 2 V and an internal resistance of $\frac{2}{3} \Omega$. The power consumption in the entire circuit is

_____ W.

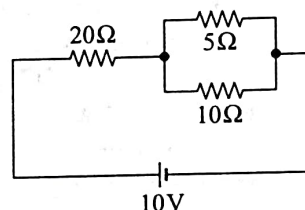
[31 Jan, 2024 (Shift-II)]

99. An electric bulb rated $50 \text{ W} - 200 \text{ V}$ is connected across a 100 V supply. The power dissipation of the bulb is:

[04 April, 2024 (Shift-II)]

- (a) 12.5 W (b) 25 W
(c) 50 W (d) 100 W

100. The ratio of heat dissipated per second through the resistance 5Ω and 10Ω in the circuit given below is: [05 April, 2024 (Shift-II)]



- (a) $1:2$ (b) $2:1$ (c) $4:1$ (d) $1:1$

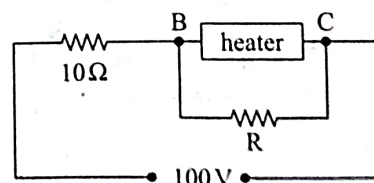
101. Water boils in an electric kettle in 20 minutes after being switched on. Using the same main supply, the length of the heating element should be to times of its initial length if the water is to be boiled in 15 minutes.

[08 April, 2024 (Shift-II)]

- (a) increased, $\frac{3}{4}$ (b) increased, $\frac{4}{3}$
(c) decreased, $\frac{3}{4}$ (d) decreased, $\frac{4}{3}$

102. A heater is designed to operate with a power of 1000 W in a 100 V line. It is connected in combination with a resistance of 10Ω and a resistance R , to a 100 V mains as shown in figure. For the heater to operate at 62.5 W , the value of R should be Ω .

[08 April, 2024 (Shift-II)]



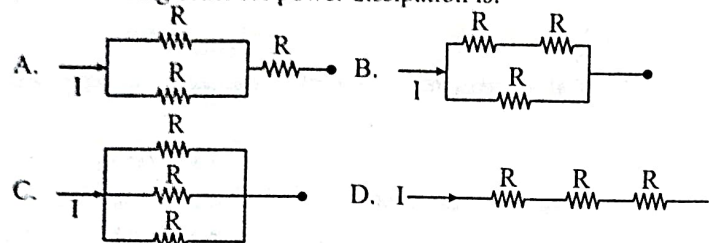
103. The H amount of thermal energy is developed by a resistor in 10 s when a current of 4 A is passed through it. If the current is increased to 16 A , the thermal energy developed by the resistor in 10 s will be:

[31 Jan, 2023 (Shift-II)]

- (a) H (b) 16H (c) $\frac{H}{4}$ (d) 4H

14. Different combination of 3 resistors of equal resistance R are shown in the figures. [13 April, 2023 (Shift-I)]

The increasing order for power dissipation is:



- (a) $P_A < P_B < P_C < P_D$ (b) $P_C < P_D < P_A < P_B$
(c) $P_B < P_C < P_D < P_A$ (d) $P_C < P_B < P_A < P_D$

15. Ratio of thermal energy released in two resistors R and $3R$ connected in parallel in an electric circuit is: [29 Jan, 2023 (Shift-I)]

- (a) 3 : 1 (b) 1 : 1 (c) 1 : 3 (d) 1 : 27

16. Two identical heater filaments are connected first in parallel and then in series. At the same applied voltage, the ratio of heat produced in same time for parallel to series will be: [11 April, 2023 (Shift-I)]

- (a) 4 : 1 (b) 2 : 1 (c) 1 : 2 (d) 1 : 4

17. A potential V_0 is applied across a uniform wire of resistance R . The power dissipation is P_1 . The wire is then cut into two equal halves and a potential of V_0 is applied across the length of each half. The total power dissipation across two wires is P_2 . The ratio $P_2 : P_1$ is $\sqrt{x} : 1$. The value of x is _____. [13 April, 2023 (Shift-I)]

18. An electrical bulb rated 220 V, 100 W, is connected in series with another bulb rated 220 V, 60 W. If the voltage across combination is 220 V, the power consumed by the 100 W bulb will be about _____ W. [28 July, 2022 (Shift-II)]

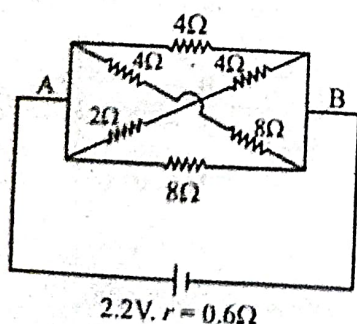
19. A resistor develops 300 J of thermal energy in 15 s, when a current of 2 A is passed through it. If the current increases to 3 A, the energy developed in 10 s is _____ J [25 June, 2022 (Shift-I)]

20. An electric bulb of 500 watt at 100 volt is used in a circuit having a 200 V supply. Calculate the resistance R to be connected in series with the bulb so that the power delivered by the bulb is 500 W. [26 Aug, 2021 (Shift-II)]

- (a) 20 Ω (b) 5 Ω (c) 10 Ω (d) 30 Ω

21. In the given figure, the emf of the cell is 2.2 V and if internal resistance is 0.6 Ω . Calculate the power dissipated in the whole circuit: [26 Aug, 2021 (Shift-I)]

- (a) 2.2 W (b) 0.65 W
(c) 4.4 W (d) 132 W



22. An electric appliance supplies 6000 J/min heat to the system. If the system delivers a power of 90 W. How long it would take to increase

the internal energy by 2.5×10^3 J? [26 Aug, 2021 (Shift-I)]

- (a) 4.1×10^1 s (b) 2.4×10^3 s
(c) 2.5×10^1 s (d) 2.5×10^2 s

113. A resistor develops 500J of thermal energy in 20 s when a current of 1.5A is passed through it. If the current is increased from 1.5 A to 3 A, what will be the energy developed in 20 s. [16 March, 2021 (Shift-II)]

- (a) 2000 J (b) 500 J
(c) 1000 J (d) 1500 J

114. A resistor dissipates 192 J of energy in 1 s when a current of 4A is passed through it. Now, when the current is doubled, the amount of thermal energy dissipated in 5 s in _____ J. [31 Aug, 2021 (Shift-II)]

115. A uniform heating wire of resistance 36 Ω is connected across a potential difference of 240 V. The wire is then cut into half and potential difference of 240 V is applied across each half separately. The ratio of power dissipation in first case to the total power dissipation in the second case would be 1 : x, where x is _____. [1 Sep, 2021 (Shift-II)]

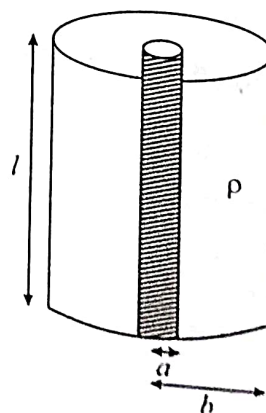
116. The energy dissipated by a resistor is 10 mJ in 1 s when an electric current of 2 mA flows through it. The resistance is _____ Ω (Round off to the Nearest Integer) [16 March, 2021 (Shift-II)]

117. An electric bulb rated as 200 W at 100 V is used in a circuit having 200 V supply. The resistance 'R' that must be put in series with the bulb so that the bulb delivers the same power is _____ Ω . [25 July, 2021 (Shift-I)]

118. In a building there are 15 bulbs of 45 W, 15 bulbs of 100 W, 15 small fans of 10 W and 2 heaters of 1kW. The voltage of electric main is 220 V. The minimum fuse capacity (rated value) of the building will be [7 Jan, 2020 (Shift-II)]

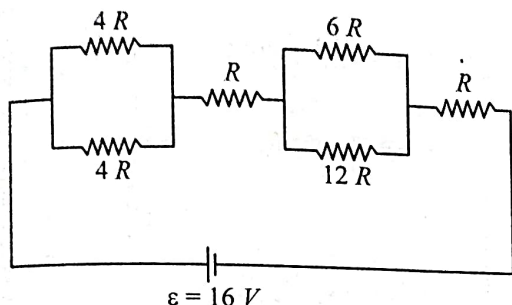
- (a) 10 A (b) 20 A
(c) 25 A (d) 15 A

119. Model a torch battery of length l to be made up of a thin cylindrical bar of radius 'a' and a concentric thin cylindrical shell of radius 'b' filled in between with an electrolyte of resistivity ρ (see figure). If the battery is connected to a resistance of value R , the maximum Joule heating in R will take place for [3 Sep, 2020 (Shift-I)]



- (a) $R = \frac{\rho}{\pi l} \ln\left(\frac{b}{a}\right)$ (b) $R = \frac{2\rho}{\pi l} \ln\left(\frac{b}{a}\right)$
(c) $R = \frac{\rho}{2\pi l} \ln\left(\frac{b}{a}\right)$ (d) $R = \frac{\rho}{2\pi l} \ln\left(\frac{a}{b}\right)$

120. A battery of 3.0 V is connected to a resistor dissipating 0.5 W of power. If the terminal voltage of the battery is 2.5 V, the power dissipated within the internal resistance is [4 Sep, 2020 (Shift-I)]
 (a) 0.10 W (b) 0.072 W
 (c) 0.50 W (d) 0.125 W
121. The resistive network shown below is connected to a D.C. source of 16 V. The power consumed by the network is 4 Watt. The value of R is: [12 April, 2019 (Shift-I)]



- (a) 8Ω (b) 6Ω
 (c) 16Ω (d) 1Ω
122. Two electric bulbs, rated at (25 W, 220 V) and (100 W, 220 V), are connected in series across a 220 V voltage source. If the 25 W and 100 W bulbs draw powers P_1 and P_2 respectively, then: [12 Jan, 2019 (Shift-I)]
 (a) $P_1 = 16$ W, $P_2 = 4$ W (b) $P_1 = 16$ W, $P_2 = 9$ W
 (c) $P_1 = 9$ W, $P_2 = 16$ W (d) $P_1 = 4$ W, $P_2 = 16$ W
123. A cell of internal resistance r drives current through an external resistance R . The power delivered by the cell to the external resistance will be maximum when: [10 Jan, 2019 (Shift-I)]
 (a) $R = 0.001 r$ (b) $R = 1000 r$
 (c) $R = 2r$ (d) $R = r$
124. A current of 2 mA was passed through an unknown resistor which dissipated a power of 4.4 W. Dissipated power when an ideal power supply of 11 V is connected across it is: [10 Jan, 2019 (Shift-II)]
 (a) 11×10^{-5} W (b) 11×10^{-3} W
 (c) 11×10^{-4} W (d) 11×10^5 W

125. Two equal resistances when connected in series to a battery, consume electric power of 60 W. If these resistances are now connected in parallel combination to the same battery, the electric power consumed will be. [11 Jan, 2019 (Shift-I)]
 (a) 60 W (b) 240 W (c) 120 W (d) 30 W

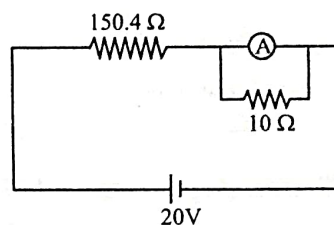
126. One kg of water, at 20°C , is heated in an electric kettle whose heating element has a mean (temperature averaged) resistance of 20Ω . The rms voltage in the mains is 200V. Ignoring heat loss from the kettle, time taken for water to evaporate fully, is close to: [Specific heat of water ≈ 4200 J/kg $^\circ\text{C}$, Latent heat of water = 2260 kJ/kg] [12 April, 2019 (Shift-II)]

- (a) 3 minutes (b) 22 minutes
 (c) 10 minutes (d) 16 minutes

127. A 2 W carbon resistor is color coded with green, black, red and brown respectively. The maximum current which can be passed through this resistor is: [10 Jan, 2019 (Shift-I)]
 (a) 20 mA (b) 100 mA
 (c) 0.4 mA (d) 63 mA

Kirchhoff's Law and Grouping of Cells

128. In the figure shown below, a resistance of 150.4Ω is connected in series to an ammeter A of resistance 240Ω . A shunt resistance of 10Ω is connected in parallel with the ammeter. The reading of the ammeter is _____ mA. [03 April, 2025 (Shift-I)]



129. Two cells of emf 1 V and 2 V and internal resistance 2Ω and 1Ω , respectively, are connected in series with an external resistance of 6Ω . The total current in the circuit is I_1 . Now the same two cells in parallel configuration are connected to same external resistance. In this case, the total current drawn is I_2 . The value of $\left(\frac{I_1}{I_2}\right)$ is $\frac{x}{3}$. The value of x is _____. [03 April, 2025 (Shift-II)]

130. Given below are two statements:

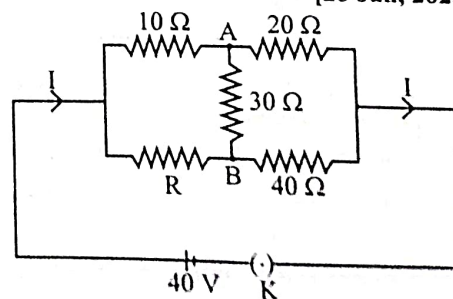
Statement-I: The equivalent emf of two nonideal batteries connected in parallel is smaller than either of the two emfs.

Statement-II: The equivalent internal resistance of two nonideal batteries connected in parallel is smaller than the internal resistance of either of the two batteries.

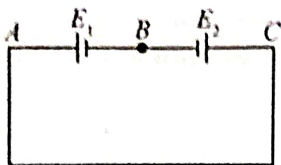
In the light of the above statements, choose the correct answer from the options given below: [22 Jan, 2025 (Shift-I)]

- (a) Both statement-I and statement-II are true.
 (b) Both statement-I and statement-II are false.
 (c) Statement-I is true but statement-II is false.
 (d) Statement-I is false but statement-II is true.

131. The value of current I in the electrical circuit as given below, when potential at A is equal to the potential at B, will be _____ A [28 Jan, 2025 (Shift-II)]

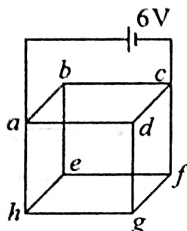


132. Two cells are connected in opposition as shown. Cell E_1 is of 8 V emf and 2Ω internal resistance; the cell E_2 is of 2 V emf and 4Ω internal resistance. The terminal potential difference of cell E_2 is: [30 Jan, 2024 (Shift-I)]



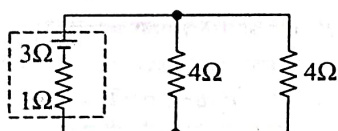
133. Twelve wires each having resistance 2Ω are joined to form a cube. A battery of 6 V emf is joined across point a and c. The voltage difference between e and f is _____ V.

[04 April, 2024 (Shift-I)]



134. In the given circuit, the terminal potential difference of the cell is:

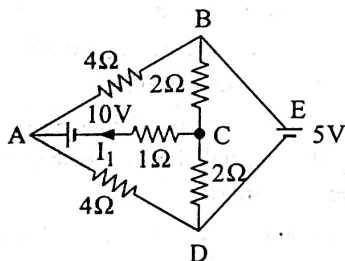
[08 April, 2024 (Shift-I)]



- (a) 2 V (b) 4 V
(c) 1.5 V (d) 3 V

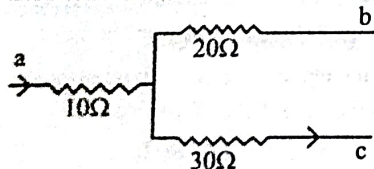
135. The current flowing through the 1Ω resistor is $n/10\text{ A}$. The value of n is _____.

[09 April, 2024 (Shift-I)]



136. Figure shows a part of an electric circuit. The potentials at points a, b and c are 30 V , 12 V and 2 V respectively. The current through the 20Ω resistor will be.

[6 April, 2023 (Shift-II)]



- (a) 0.4 A (b) 0.2 A (c) 0.6 A (d) 1.0 A

137. Two identical cells each of emf 1.5 V are connected in parallel across a parallel combination of two resistors each of resistance 20Ω . A voltmeter connected in the circuit measures 1.2 V . The internal resistance of each cell is:

[24 June, 2022 (Shift-I)]

- (a) 2.5Ω (b) 4Ω (c) 5Ω (d) 10Ω

138. Two cells of same emf but different internal resistance r_1 and r_2 are connected in series with a resistance R . The value of resistance R , for which the potential difference across second cell is zero, is:

[25 June, 2022 (Shift-II)]

- (a) $r_2 - r_1$ (b) $r_1 - r_2$
(c) r_1 (d) r_2

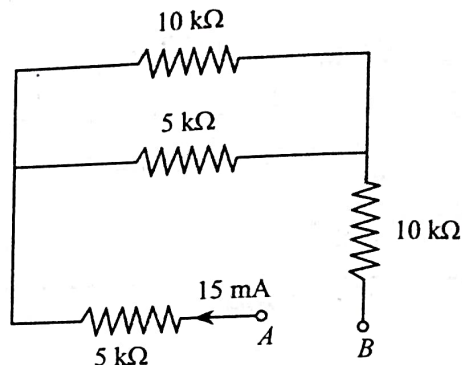
139. The combination of two identical cells, whether connected in series or parallel combination provides the same current through an external resistance of 2Ω . The value of internal resistance of each cell is:

[29 June, 2022 (Shift-I)]

- (a) 2Ω (b) 4Ω (c) 6Ω (d) 8Ω

140. A current of 15 mA flows in the circuit as shown in figure. The value of potential difference between the points A and B will be:

[25 July, 2022 (Shift-I)]



- (a) 50 V (b) 75 V (c) 150 V (d) 275 V

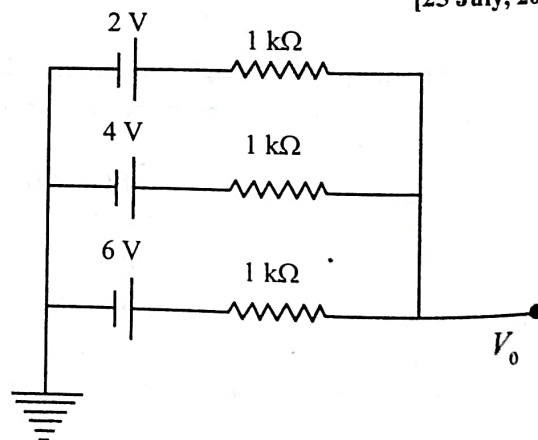
141. Two sources of equal emfs are connected in series. This combination is connected to an external resistance R . The internal resistances of the two sources are r_1 and r_2 ($r_1 > r_2$). If the potential difference across the source of internal resistance r_1 is zero, then the value of R will be

[27 July, 2022 (Shift-I)]

- (a) $r_1 - r_2$ (b) $\frac{r_1 r_2}{r_1 + r_2}$
(c) $\frac{r_1 + r_2}{2}$ (d) $r_2 - r_1$

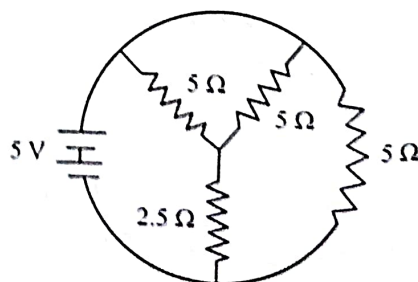
142. In the given figure, the value of V_0 will be _____ V.

[25 July, 2022 (Shift-I)]

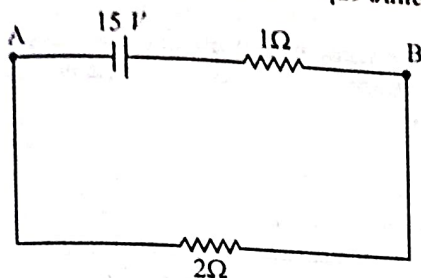


143. The total current supplied to the circuit as shown in figure by the 5 V battery is _____ A.

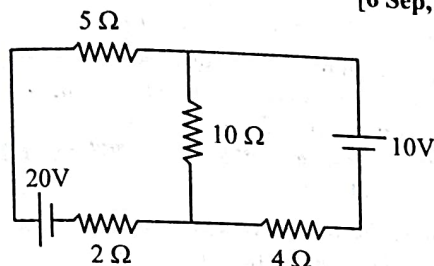
[25 June, 2022 (Shift-I)]



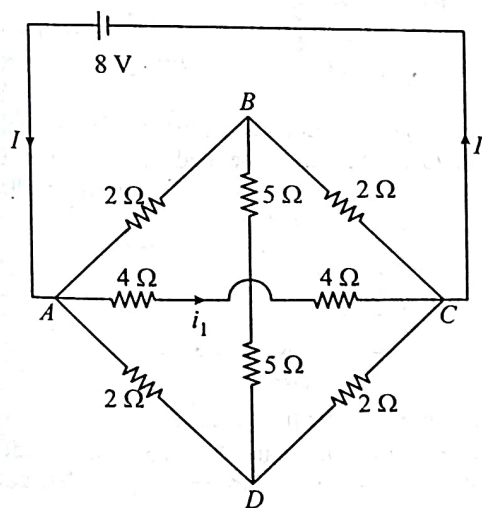
144. For the network shown below, the value of $V_B - V_A$ is _____
[29 June, 2022 (Shift-I)]



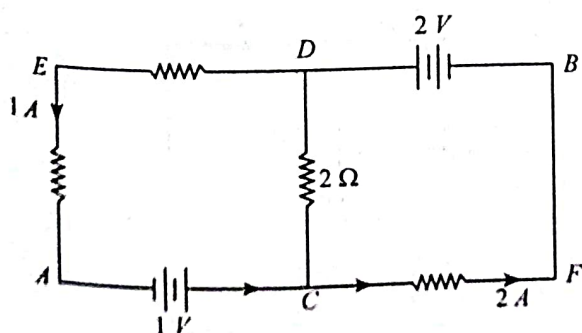
145. In the figure shown, the current in the 10 V battery is close to:
[6 Sep, 2020 (Shift-II)]



- (a) 0.71 A from positive to negative terminal
(b) 0.42 A from positive to negative terminal
(c) 0.21 A from positive to negative terminal.
(d) 0.36 A from negative to positive terminal.
146. The value of current i_1 flowing from A to C in the circuit diagram is
[4 Sep, 2020 (Shift-II)]

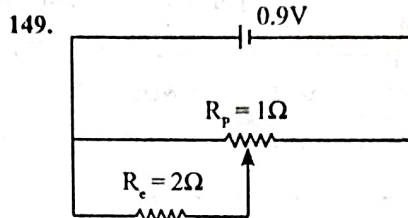


- (a) 1 A (b) 4 A (c) 5 A (d) 2 A
147. In the circuit, given in the figure currents in different branches and value of one resistor are shown. Then potential at point B with respect to the point A is:
[5 Sep, 2020 (Shift-II)]



148. The series combination of two batteries, both of the same emf 10 V, but different internal resistance of 20Ω and 5Ω , is connected to the parallel combination of two resistors 30Ω and $R\Omega$. The voltage difference across the battery of internal resistance 20Ω is zero, the value of R (in Ω) is _____.
[8 Jan, 2020 (Shift-II)]

Current Measuring Instrument



149.

Sliding contact of a potentiometer is in the middle of the potentiometer wire having resistance $R_p = 1\Omega$ as shown in the figure. An external resistance of $R_c = 2\Omega$ is connected via the sliding contact.

[22 Jan, 2025 (Shift-I)]

The electric current in the circuit is:

- (a) 0.9 A (b) 1.0 A (c) 0.3 A (d) 1.35 A
150. A galvanometer having a coil of resistance 30Ω need 20 mA of current for full-scale deflection. If a maximum current of 3 A is to be measured using this galvanometer, the resistance of the shunt to be added to the galvanometer should be $\frac{30}{X}\Omega$, where X is

[23 Jan, 2025 (Shift-II)]

- (a) 596 (b) 298 (c) 447 (d) 149
151. A wire of length 10 cm and radius $\sqrt{7} \times 10^{-4}\text{m}$ connected across the right gap of a meter bridge. When a resistance of 4.5Ω is connected on the left gap by using a resistance box, the balance length is found to be at 60 cm from the left end. If the resistivity of the wire is $R \times 10^{-7}\Omega\text{m}$, then value of R is:
[27 Jan, 2024 (Shift-I)]

- (a) 63 (b) 70 (c) 66 (d) 35
152. Wheatstone bridge principle is used to measure the specific resistance (S_1) of given wire, having length L , radius r . If X is the resistance of wire, then specific resistance is: $S_1 = X \left(\frac{\pi r^2}{L} \right)$. If

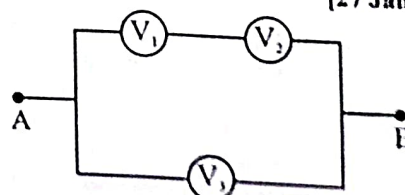
the length of the wire gets doubled then the value of specific resistance will be:

[27 Jan, 2024 (Shift-II)]

- (a) $\frac{S_1}{4}$ (b) $2S_1$ (c) $\frac{S_1}{2}$ (d) S_1

153. Three voltmeters, all having different internal resistances are joined as shown in figure. When some potential difference is applied across A and B, their readings are V_1 , V_2 and V_3 . Choose the correct option.

[27 Jan, 2024 (Shift-II)]



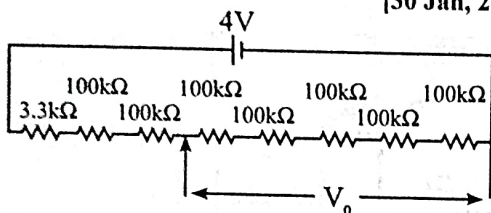
[27 Jan, 2024 (Shift-II)]

- (a) $V_1 = V_2$ (b) $V_1 \neq V_2 = V_3$
 (c) $V_1 + V_2 > V_3$ (d) $V_1 + V_2 = V_3$

154. A galvanometer having coil resistance 10Ω shows a full scale deflection for a current of 3mA . For it to measure a current of 8A , the value of the shunt should be: [29 Jan, 2024 (Shift-I)]

- (a) $3 \times 10^{-3}\Omega$ (b) $4.85 \times 10^{-3}\Omega$
 (c) $3.75 \times 10^{-3}\Omega$ (d) $2.75 \times 10^{-3}\Omega$

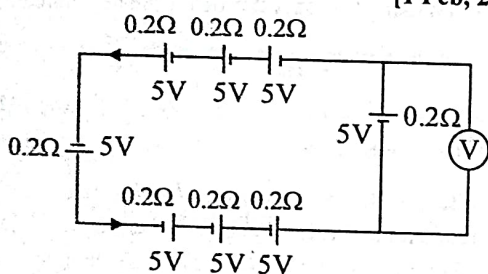
155. A potential divider circuit is shown in figure. The output voltage V_0 is [30 Jan, 2024 (Shift-I)]



- (a) 4V (b) 2mV
 (c) 0.5V (d) 12mV

156. Two resistance of 100Ω and 200Ω are connected in series with a battery of 4V and negligible internal resistance. A voltmeter is used to measure voltage across 100Ω resistance, which gives reading as 1V . The resistance of voltmeter must be Ω . [30 Jan, 2024 (Shift-II)]

157. The reading in the ideal voltmeter (V) shown in the given circuit diagram is: [1 Feb, 2024 (Shift-I)]



- (a) 5V (b) 10V (c) 0V (d) 3V

158. A galvanometer has a resistance of 50Ω and it allows maximum current of 5mA . It can be converted into voltmeter to measure upto 100V by connecting in series a resistor of resistance [1 Feb, 2024 (Shift-I)]

- (a) 5975Ω (b) 20050Ω
 (c) 19950Ω (d) 19500Ω

159. The resistance per centimeter of a meter bridge wire is r , with $X\Omega$ resistance in left gap. Balancing length from left end is at 40cm with 25Ω resistance in right gap. Now the wire is replaced by another wire of $2r$ resistance per centimeter. The new balancing length for same settings will be at [31 Jan, 2024 (Shift-II)]

- (a) 20cm (b) 10cm (c) 80cm (d) 40cm

160. In an ammeter, 5% of the main current passes through the galvanometer. If resistance of the galvanometer is G , the resistance of ammeter will be: [1 Feb, 2024 (Shift-II)]

- (a) $\frac{G}{200}$ (b) $\frac{G}{199}$ (c) $199G$ (d) $200G$

161. To measure the temperature coefficient of resistivity α of a semiconductor, an electrical arrangement shown in the figure is prepared. The arm BC is made up of the semiconductor. The experiment is being conducted at 25°C and resistance of the semiconductor arm is $3\text{m}\Omega$. Arm BC is cooled at a constant rate of 2°C/s . If the galvanometer G shows no deflection after 10s , then α is: [1 Feb, 2024 (Shift-II)]

- (a) $-2 \times 10^{-2}^\circ\text{C}^{-1}$ (b) $-1.5 \times 10^{-2}^\circ\text{C}^{-1}$
 (c) $-1 \times 10^{-2}^\circ\text{C}^{-1}$ (d) $-2.5 \times 10^{-2}^\circ\text{C}^{-1}$

162. In a metre-bridge when a resistance in the left gap is 2Ω and unknown resistance in the right gap, the balance length is found to be 40cm . On shunting the unknown resistance with 2Ω , the balance length changes by: [1 Feb, 2024 (Shift-II)]

- (a) 22.5cm (b) 20cm (c) 62.5cm (d) 65cm

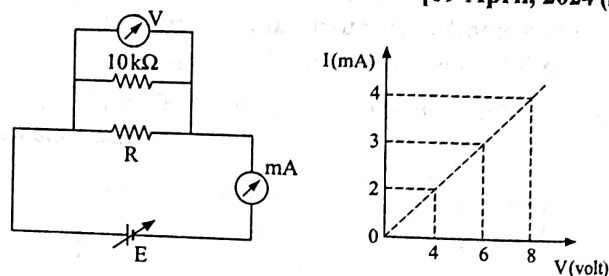
163. To measure the internal resistance of a battery, potentiometer is used. For $R = 10\Omega$, the balance point is observed at $\ell = 500\text{cm}$ and for $R = 1\Omega$ the balance point is observed at $\ell = 400\text{cm}$. The internal resistance of the battery is approximately: [04 April, 2024 (Shift-I)]

- (a) 0.2Ω (b) 0.4Ω (c) 0.1Ω (d) 0.3Ω

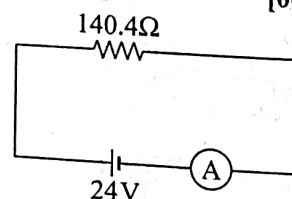
164. A galvanometer of resistance 100Ω when connected in series with 400Ω measures a voltage of upto 10V . The value of resistance required to convert the galvanometer into ammeter to read upto 10A is $x \times 10^{-2}\Omega$. The value of x is: [05 April, 2024 (Shift-II)]

- (a) 2 (b) 800 (c) 20 (d) 200

165. To determine the resistance (R) of a wire, a circuit is designed below. The V - I characteristic curve for this circuit is plotted for the voltmeter and the ammeter readings as shown in figure. The value of R is Ω . [09 April, 2024 (Shift-II)]



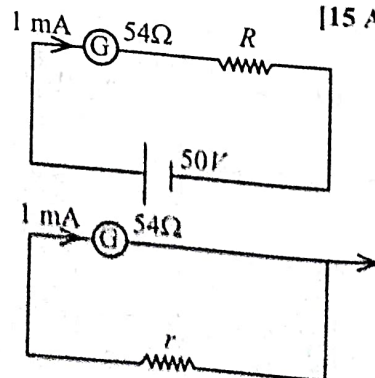
166. In the given figure an ammeter A consists of a 240Ω coil connected in parallel to a 10Ω shunt. The reading of the ammeter is mA . [06 April, 2024 (Shift-II)]



167. A galvanometer has a coil of resistance 200Ω with a full scale deflection at $20\mu\text{A}$. The value of resistance to be added to use it as an ammeter of range $(0-20)\text{mA}$ is: [09 April, 2024 (Shift-II)]

- (a) 0.40Ω (b) 0.20Ω
 (c) 0.50Ω (d) 0.10Ω

168. For designing a voltmeter of range 50V and an ammeter of range 10mA using a galvanometer which has a coil of resistance 54Ω showing a full scale deflection for 1mA as in figure. [15 April, 2023 (Shift-II)]



- (A) for voltmeter $R \approx 50 \text{ k}\Omega$ (B) for ammeter $r \approx 0.2 \Omega$
 (C) for ammeter $r \approx 6 \Omega$ (D) for voltmeter $R \approx 5 \text{ k}\Omega$
 (E) for voltmeter $R \approx 500 \Omega$

Choose the correct answer from the options given below:

- (a) (C) and (E) (b) (C) and (D)
 (c) (A) and (C) (d) (A) and (B)

169. Given below are two statements: One is labelled as Assertion (A) and the other is labelled as Reason (R).

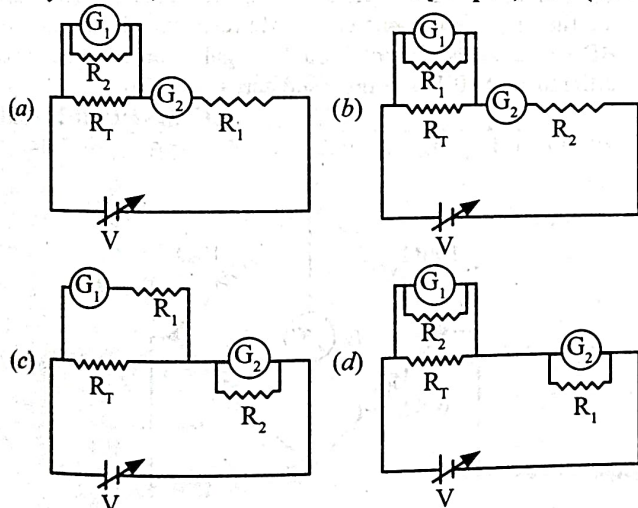
Assertion (A): For measuring the potential difference across a resistance of 600Ω , the voltmeter with resistance 1000Ω will be preferred over voltmeter with resistance 4000Ω .

Reason (R): Voltmeter with higher resistance will draw smaller current than voltmeter with lower resistance.

In the light of the above statements, choose the most appropriate answer from the options given below. [1 Feb, 2023 (Shift-II)]

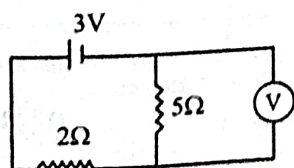
- (a) (A) is not correct but (R) is correct
 (b) Both (A) and (R) are correct and (R) is the correct explanation of (A)
 (c) Both (A) and (R) are correct but (R) is not the correct explanation of (A)
 (d) (A) is correct but R is not correct

170. A student is provided with a variable voltage source V , a test resistor $R_T = 10 \Omega$, two identical galvanometers G_1 and G_2 and two additional resistors, $R_1 = 10 \text{ M}\Omega$ and $R_2 = 0.001 \Omega$. For conducting an experiment to verify ohms law, the most suitable circuit is: [6 April, 2023 (Shift-II)]



171. When two resistances R_1 and R_2 are connected in series and introduced into the left gap of a meter bridge and a resistance of 10Ω is introduced into the right gap, a null point is found at 60 cm from the left side. When R_1 and R_2 are connected in parallel and introduced into the left gap, a resistance of 3Ω is introduced into the right-gap to get null point at 40 cm from left the end. The product of R_1, R_2 is Ω^2 [29 Jan, 2023 (Shift-II)]

172. As shown in the figure, the voltmeter reads 2 V across 5Ω resistor. The resistance of the voltmeter is Ω . [6 April, 2023 (Shift-II)]



173. When a resistance of 5Ω is shunted with a moving coil galvanometer, it shows a full scale deflection for a current of 250 mA , however when 1050Ω resistance is connected with it in series, it gives full scale deflection for 25 volt . The resistance of galvanometer is Ω . [13 April, 2023 (Shift-I)]

174. In a metre bridge experiment the balance point is obtained if the gaps are closed by 2Ω and 3Ω . A shunt of $X \Omega$ is added to 3Ω resistor to shift the balancing point by 22.5 cm . The value of X is Ω . [29 Jan, 2023 (Shift-I)]

175. A null point is found at 200 cm in potentiometer when cell in secondary circuit is shunted by 5Ω . When a resistance of 15Ω is used for shunting null point moves to 300 cm . The internal resistance of the cell is Ω . [29 Jan, 2023 (Shift-II)]

176. A 72Ω galvanometer is shunted by a resistance of 8Ω . The percentage of the total current which passes through the galvanometer is: [27 June, 2022 (Shift-I)]

- (a) 0.1% (b) 10%
 (c) 25% (d) 0.25%

177. A teacher in his physics laboratory allotted an experiment to determine the resistance (G) of a galvanometer. Students took the observations for $\frac{1}{3}$ deflection in the galvanometer. Which of the

below is true for measuring value of G ? [25 June, 2022 (Shift-I)]

- (a) $\frac{1}{3}$ deflection method cannot be used for determining the resistance of the galvanometer.
 (b) $\frac{1}{3}$ deflection method can be used and in this case the G equals to twice the value of shunt resistance(s).
 (c) $\frac{1}{3}$ deflection method can be used and in this case, the G equals to three times the value of shunt resistance(s).
 (d) $\frac{1}{3}$ deflection method can be used and in this case, the G value equals to the shunt resistance(s).

178. A 72Ω galvanometer is shunted by a resistance of 8Ω . The percentage of the total current which passes through the galvanometer is: [27 June, 2022 (Shift-I)]

- (a) 0.1% (b) 10%
 (c) 25% (d) 0.25%

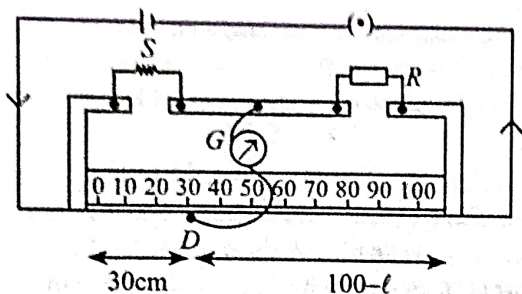
179. If n represents the actual number of deflections in a converted galvanometer of resistance G and shunt resistance S . Then the total current I when its figure of merit is K will be

[25 June, 2022 (Shift-II)]

- (a) $\frac{KS}{(S+G)}$ (b) $\frac{(G+S)}{nKS}$
 (c) $\frac{nKS}{(G+S)}$ (d) $\frac{nK(G+S)}{S}$

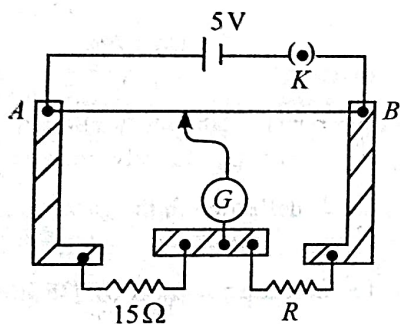
180. In a meter bridge experiment, for measuring unknown resistance ' S ', the null point is obtained at a distance 30 cm from the left side as shown at point D . If R is $5.6 \text{ k}\Omega$, then the value of unknown resistance ' S ' will be Ω .

[27 July, 2022 (Shift-I)]



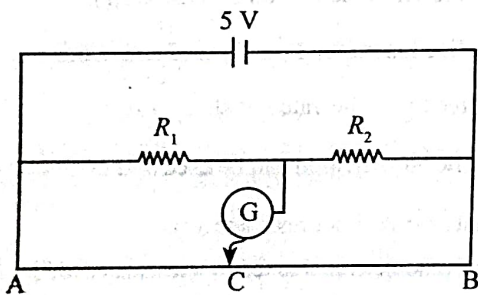
181. A meter bridge setup is shown in the figure. It is used to determine an unknown resistance R using a given resistor of 15Ω . The galvanometer (G) shows null deflection when tapping key is at 43 cm mark from end A . If the end correction for end A is 2 cm. Then the determined value of R will be _____ Ω .

[28 June, 2022 (Shift-I)]



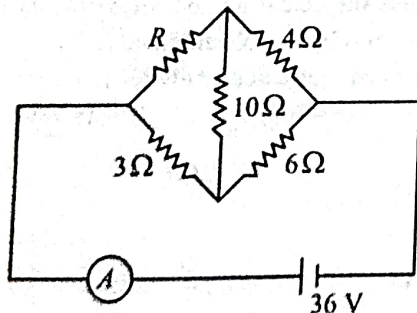
182. In the given figure of meter bridge experiment, the balancing length AC corresponding to null deflection of the galvanometer is 40 cm. The balancing length, if the radius of the wire AB is doubled, will be _____ cm.

[27 July, 2022 (Shift-II)]



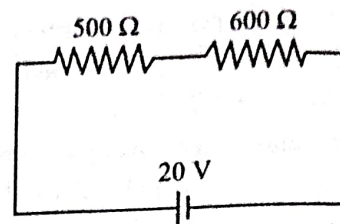
183. Current measured by the ammeter (A) in the reported circuit when no current flows through 10 resistance, will be _____ A.

[28 June, 2022 (Shift-I)]



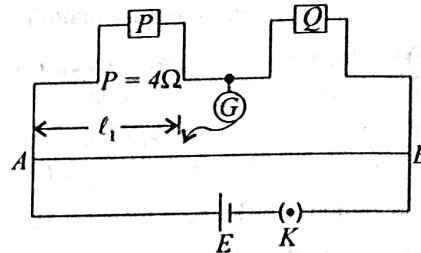
184. Two resistors are connected in series across a battery as shown in figure. If a voltmeter of resistance 2000Ω is used to measure the potential difference across 500Ω resistor, the reading of the voltmeter will be _____ V.

[29 June, 2022 (Shift-II)]



185. Resistances are connected in a meter bridge circuit as shown in the figure. The balancing length ℓ_2 is 40 cm. Now an unknown resistance x is connected in series with P and new balancing length is found to be 80 cm measured from the same end. Then the value of x will be _____ Ω .

[26 July, 2022 (Shift-I)]



186. For full scale deflection of total 50 divisions, 50 mV voltage is required in galvanometer. The resistance of galvanometer if its current sensitivity is 2 div/mA will be: [27 Aug, 2021 (Shift-II)]

(a) 2Ω (b) 4Ω (c) 1Ω (d) 5Ω

187. Consider a galvanometer shunted with 5Ω resistance and 2% of current passes through it. What is the resistance of the given galvanometer?

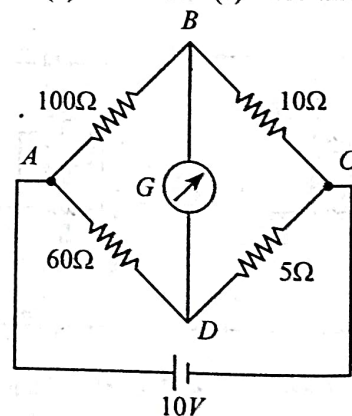
[31 Aug, 2021 (Shift-I)]

(a) 300Ω (b) 344Ω (c) 226Ω (d) 245Ω

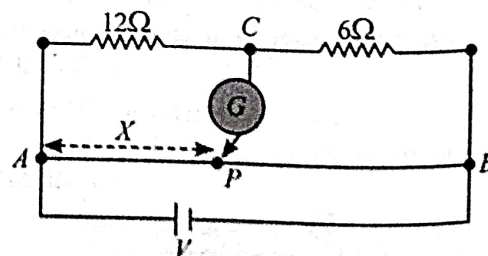
188. The four arms of a Wheatstone bridge have resistances as shown in the figure. A galvanometer of 15Ω resistance is connected across BD . Calculate the current through the galvanometer when a potential difference of $10V$ is maintained across AC .

[17 March, 2021 (Shift-II)]

(a) $4.87\mu A$ (b) $2.44mA$ (c) $4.87mA$ (d) $2.44\mu A$



189. Consider a 72 cm long wire AB as shown in the figure. The galvanometer jockey is placed at P on AB at a distance x cm from A . The galvanometer shows zero deflection.



The value of x , to nearest integer, is _____

[18 March, 2021 (Shift-II)]

A galvanometer of resistance G is converted into a voltmeter of range $0 - 1\text{ V}$ by connecting a resistance R_1 in series with it. The additional resistance that should be connected in series with R_1 to increase the range of the voltmeter to $0 - 2\text{ V}$ will be:

[5 Sep, 2020 (Shift-I)]

- (a) $R_1 + G$ (c) G (b) R_1 (d) $R_1 - G$
91. Two resistors 400Ω and 800Ω are connected in series across a 6 V battery. The potential difference measured by a voltmeter of $10\text{ k}\Omega$ across 400Ω resistor is close to:

[3 Sep, 2020 (Shift-II)]

- (a) 2 V (b) 2.05 V (c) 1.95 V (d) 1.8 V
92. A galvanometer is used in laboratory for detecting the null point in electrical experiments. If, on passing a current of 6 mA it produces a deflection of 2° , its figure of merit is close to:

[5 Sep, 2020 (Shift-II)]

- (a) 666° A/div . (b) $6 \times 10^{-3}\text{ A/div}$
(c) 333° A/div . (d) $3 \times 10^{-3}\text{ A/div}$

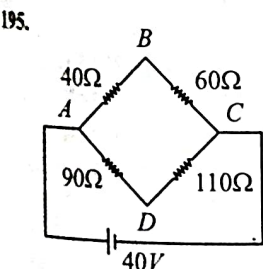
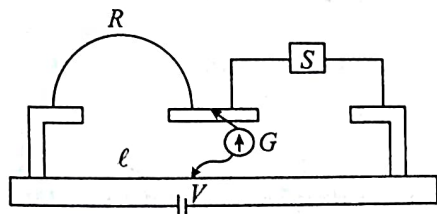
93. A galvanometer having a coil resistance 100Ω gives a full scale deflection when a current of 1 mA is passed through it. What is the value of the resistance which can convert this galvanometer into a voltmeter giving full scale deflection for a potential difference of 10 V .

[8 Jan, 2020 (Shift-II)]

- (a) $9.9\text{ k}\Omega$ (b) $7.9\text{ k}\Omega$ (c) $10\text{ k}\Omega$ (d) $8.9\text{ k}\Omega$

94. In a meter bridge experiment S is a standard resistance. R is a resistance wire. It is found that balancing length is $\ell = 25\text{ cm}$. If R is replaced by a wire of half-length and half diameter that of R of same material, then the balancing distance ℓ (in cm) will now be

[9 Jan, 2020 (Shift-II)]



Four resistance 40Ω , 60Ω , 90Ω , and 110Ω , make the arms of a quadrilateral $ABCD$. Across AC is a battery of emf 40 V and internal resistance negligible. The potential difference across BD in V is

[4 Sep, 2020 (Shift-II)]

96. The resistance of a galvanometer is 50 ohm and the maximum current which can be passed through it is 0.002 A . What resistance must be connected to it in order to convert it into an ammeter of range $0 - 0.5\text{ A}$?

[9 April, 2019 (Shift-II)]

- (a) 0.2 ohm (b) 0.002 ohm
(c) 0.02 ohm (d) 0.5 ohm

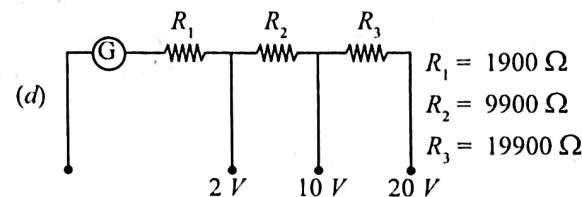
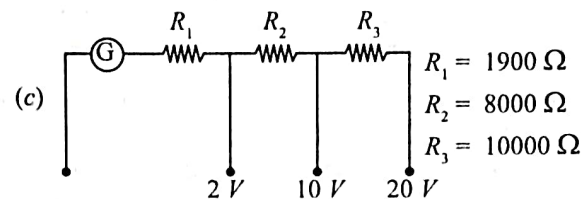
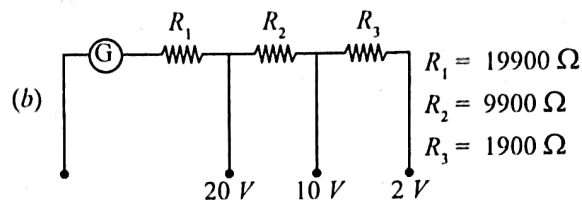
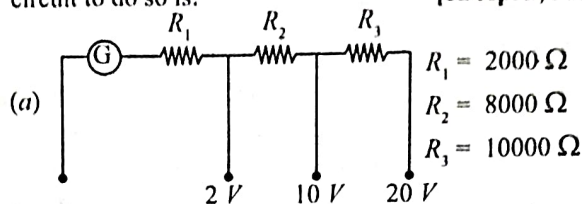
97. A moving coil galvanometer has resistance 50Ω and it indicates full deflection at 4 mA current. A voltmeter is made using this galvanometer and a $5\text{ k}\Omega$ resistance. The maximum voltage, that can be measured using this voltmeter, will be close to:

[9 April, 2019 (Shift-I)]

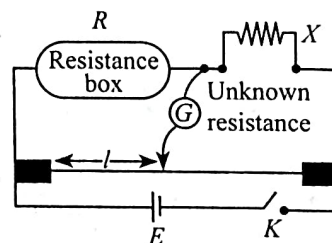
- (a) 15 V (b) 20 V (c) 10 V (d) 40 V

198. A galvanometer of resistance 100Ω has 50 divisions on its scale and has sensitivity of $20\mu\text{A/division}$. It is to be converted to a voltmeter with three ranges, of $0 - 2\text{ V}$, $0 - 10\text{ V}$ and $0 - 20\text{ V}$. The appropriate circuit to do so is:

[12 April, 2019 (Shift-I)]



199. In a meter bridge experiment, the circuit diagram and the corresponding observation table are shown in figure.



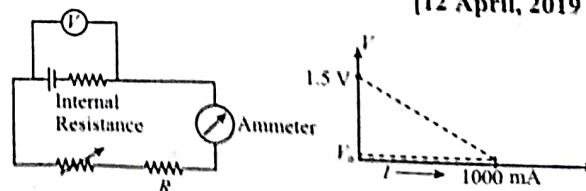
Sl. No.	$R (\Omega)$	$l (\text{cm})$
1.	1000	60
2.	100	13
3.	10	1.5
4.	1	1.0

Which of the readings is inconsistent? [10 April, 2019 (Shift-I)]

- (a) 4 (b) 3 (c) 2 (d) 1

200. To verify Ohm's law, a student connects the voltmeter across the battery as, shown in the figure. The measured voltage is plotted as a function of the current, and the following graph is obtained. If V_0 is almost zero, identify the correct statement

[12 April, 2019 (Shift-I)]



- (a) The value of the resistance R is 1.5Ω
 (b) The emf of the battery is 1.5 V and the value of R is 1.5Ω
 (c) The emf of the battery is 1.5 V and its internal resistance is 1.5Ω
 (d) The potential difference across the battery is 1.5 V when it sends a current of 1000 mA

201. An ideal battery of 4 V and resistance R are connected in series in the primary circuit of a potentiometer of length 1 m and resistance 5Ω . The value of R , to give a difference of 5 mV across 10 cm of potentiometer wire, is: [12 Jan, 2019 (Shift-I)]

(a) 490Ω (b) 480Ω (c) 395Ω (d) 495Ω

202. A moving coil galvanometer allows a full scale current of 10^{-4} A . A series resistance of $2 \text{ M}\Omega$ is required to convert the above galvanometer into a voltmeter of range $0 - 5 \text{ V}$. Therefore the value of shunt resistance required to convert the above galvanometer into an ammeter of range $0 - 10 \text{ mA}$ is: [10 April, 2019 (Shift-I)]

(a) 200Ω (b) 100Ω (c) 10Ω (d) 500Ω

203. A moving coil galvanometer, having a resistance G , produces full scale deflection when a current I_g flows through it. This galvanometer can be converted into (i) an ammeter of range 0 to I_0 ($I_0 > I_g$) by connecting a shunt resistance R_A to it and (ii) into a voltmeter of range 0 to V ($V = GI_0$) by connecting a series resistance R_V to it. Then, [12 April, 2019 (Shift-II)]

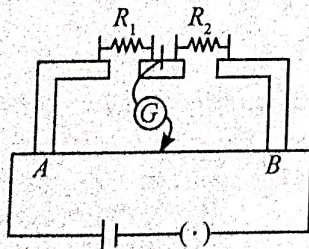
$$(a) R_A R_V = G^2 \left(\frac{I_g}{I_0 - I_g} \right) \text{ and } \frac{R_A}{R_V} = \left(\frac{I_0 - I_g}{I_g} \right)^2$$

$$(b) R_A R_V = G^2 \text{ and } \frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$$

$$(c) R_A R_V = G^2 \text{ and } \frac{R_A}{R_V} = \frac{I_g}{I_0 - I_g}$$

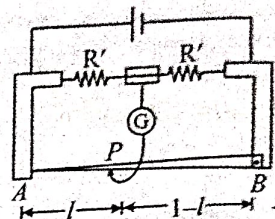
$$(d) R_A R_V = G^2 \left(\frac{I_0 - I_g}{I_g} \right) \text{ and } \frac{R_A}{R_V} = \left(\frac{I_g}{I_0 - I_g} \right)^2$$

204. In the experimental setup of metre bridge shown in the figure, the null point is obtained at a distance of 40 cm from A . If a 10Ω resistor is connected in series with R_1 , the null point shifts by 10 cm . The resistance that should be connected in parallel with $(R_1 + 10) \Omega$ such that the null point shifts back to its initial position is _____:



(a) 20Ω (b) 40Ω (c) 60Ω (d) 30Ω

205. In a meter bridge, the wire of length 1 m has a non-uniform cross-section such that the variation $\frac{dR}{dl}$ of its resistance R with length l is $\frac{dR}{dl} \propto \frac{1}{\sqrt{l}}$. Two equal resistances are connected as shown in the figure. The galvanometer has zero deflection when the jockey is at point P . What is the length AP ? [12 Jan, 2019 (Shift-I)]



(a) 0.2 m (b) 0.3 m (c) 0.25 m (d) 0.35 m

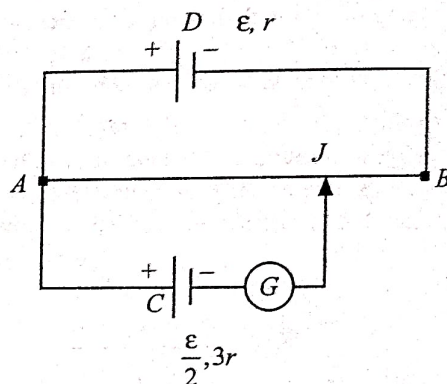
206. A galvanometer, whose resistance is 50 ohm , has 25 divisions. When a current of $4 \times 10^{-4} \text{ A}$ passes through it, its needle (pointer) deflects by one division. To use this galvanometer as a voltmeter of range 2.5 V , it should be connected to a resistance of:

[12 Jan, 2019 (Shift-I)]

(a) 250 ohm (b) 200 ohm (c) 6200 ohm (d) 6250 ohm

207. A potentiometer wire AB having length L and resistance $12r$ is joined to a cell D of emf ϵ and internal resistance r . A cell C having emf $\epsilon/2$ and internal resistance $3r$ is connected. The length AJ at which the galvanometer as shown in figure shows no deflection is:

[10 Jan, 2019 (Shift-I)]



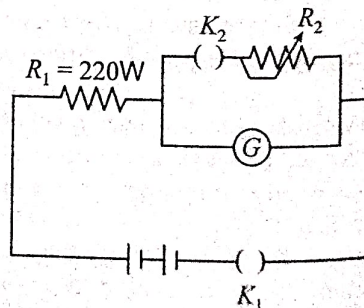
(a) $\frac{11}{12} L$ (b) $\frac{11}{24} L$ (c) $\frac{13}{24} L$ (d) $\frac{5}{12} L$

208. A galvanometer having a resistance of 20Ω and 30 divisions on both sides has figure of merit $0.005 \text{ ampere/division}$. The resistance that should be connected in series such that it can be used as a voltmeter upto 15 volt is:

[11 Jan, 2019 (Shift-I)]

(a) 100Ω (b) 120Ω (c) 80Ω (d) 125Ω

209. The galvanometer deflection, when key K_1 is closed but K_2 is open, equals θ_0 (see figure). On closing K_2 also and adjusting R_2 to 5Ω , the deflection in galvanometer becomes $\frac{\theta_0}{5}$. The resistance of the galvanometer is, then, given by [Neglect the internal resistance of battery]: [12 Jan, 2019 (Shift-I)]



(a) 5Ω (b) 22Ω (c) 25Ω (d) 12Ω

210. In a Wheatstone bridge (see figure) Resistance P and Q are approximately equal. When $R = 400 \Omega$, the bridge is balanced. On interchanging P and Q , the value of R , for balance is 405Ω . The value of X is close to:

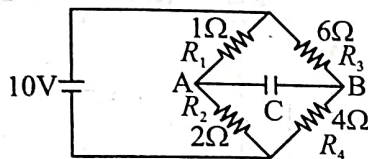
[11 Jan, 2019 (Shift-I)]

(a) 401.5 ohm (b) 404.5 ohm (c) 403.5 ohm (d) 402.5 ohm

211. The resistance of the meter bridge AB in given figure is 4Ω . With a cell of emf $\varepsilon = 0.05\text{ V}$ and rheostat resistance $R_h = 2\Omega$ the null point is obtained at some point J . When the cell is replaced by another one of emf $\varepsilon = \varepsilon_2$ the same null point J is found for $R_h = 6\Omega$. The emf ε_2 is:
 (a) 0.4 V (b) 0.3 V (c) 0.6 V (d) 0.5 V [11 Jan, 2019 (Shift-I)]

RC Circuit

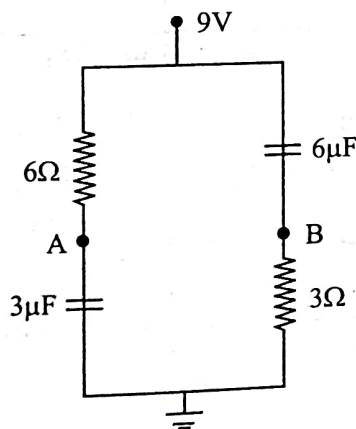
212. The charge accumulated on the capacitor connected in the following circuit is μC
 (Given $C = 150\text{ }\mu\text{F}$)



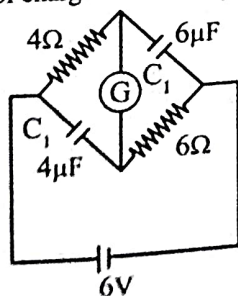
[27 Jan, 2024 (Shift-I)]

213. A 16Ω wire is bent to form a square loop. A 9V battery with internal resistance 1Ω is connected across one of its sides. If a $4\mu\text{F}$ capacitor is connected across one of its diagonals, the energy stored by the capacitor will be $\frac{x}{2}\text{ }\mu\text{J}$. where $x = \underline{\hspace{1cm}}$. [29 Jan, 2024 (Shift-I)]

214. In the given figure, the charge stored in $6\mu\text{F}$ capacitor, when points A and B are joined by a connecting wire is μC .
 [29 Jan, 2024 (Shift-II)]



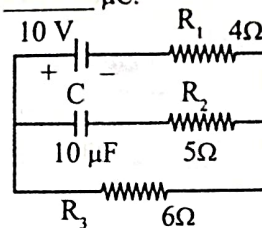
215. A galvanometer (G) of 2Ω resistance is connected in the given circuit. The ratio of charge stored in C_1 and C_2 is:



[1 Feb, 2024 (Shift-II)]

- (a) $\frac{2}{3}$ (b) $\frac{3}{2}$ (c) 1 (d) $\frac{1}{2}$

216. In an electrical circuit drawn below the amount of charge stored in the capacitor is μC .

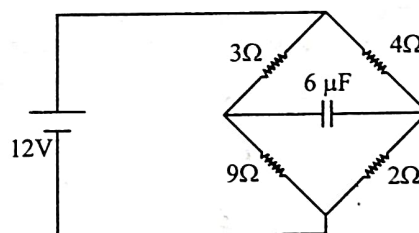


[1 Feb, 2024 (Shift-II)]

217. The electric field between the two parallel plates of a capacitor of $1.5\text{ }\mu\text{F}$ capacitance drops to one third of its initial value in $6.6\text{ }\mu\text{s}$ when the plates are connected by a thin wire. The resistance of this wire is Ω . (Given, $\log 3 = 1.1$)

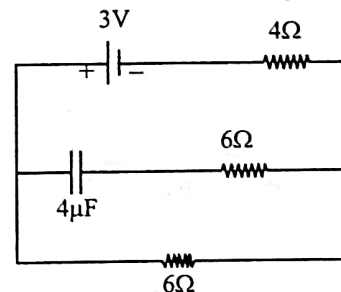
[05 April, 2024 (Shift-I)]

218. In the circuit shown, the energy stored in the capacitor is $n\text{ }\mu\text{J}$. The value of n is $\underline{\hspace{1cm}}$.
 [13 April, 2023 (Shift-II)]



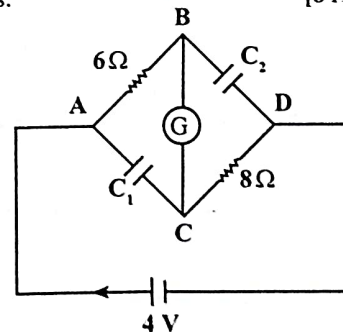
219. In the network shown below, the charge accumulated in the capacitor in steady state will be:

[13 April, 2023 (Shift-II)]



- (a) $7.2\text{ }\mu\text{C}$ (b) $4.8\text{ }\mu\text{C}$ (c) $10.3\text{ }\mu\text{C}$ (d) $12\text{ }\mu\text{C}$

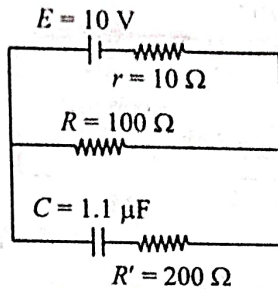
220. In this figure the resistance of the coil of galvanometer G is 2Ω . The emf of the cell is 4 V . The ratio of potential difference across C_1 and C_2 is:
 [8 April, 2023 (Shift-I)]



- (a) 1 (b) $\frac{4}{5}$ (c) $\frac{3}{4}$ (d) $\frac{5}{4}$

221. A capacitor is discharging through a resistor R . Consider in time t_1 the energy stored in the capacitor reduces to half of its initial value and in time t_2 , the charge stored in the capacitor reduces to half of its initial value and in time t_3 , the charge stored reduces to one eighth of its initial value. The ratio t_1/t_2 will be [29 June, 2022 (Shift-II)]
 (a) $1/2$ (b) $1/3$ (c) $1/4$ (d) $1/6$

222. As shown in the figure, in steady state, the charge stored in the capacitor is $\times 10^{-6} \text{C}$.
[27 July, 2022 (Shift-II)]



223. A capacitor is connected to a 20 V battery through a resistance of 10Ω . It is found that the potential difference across the capacitor rises to 2 V in $1 \mu\text{s}$. The capacitance of the capacitor is μF

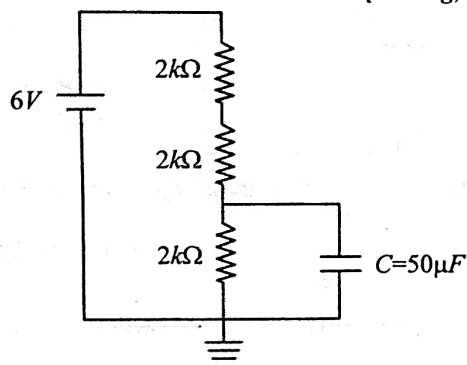
Given $\ln\left(\frac{10}{9}\right) = 0.105$

[1 Sep, 2021 (Shift-II)]

- (a) 1.85 (b) 0.95 (c) 0.105 (d) 9.52

224. A capacitor of $50 \mu\text{F}$ is connected in a circuit as shown in figure. The charge on the upper plate of the capacitor is μC .

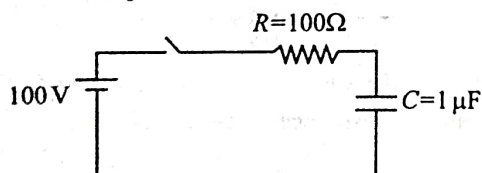
[31 Aug, 2021 (Shift-I)]



225. A capacitor of capacitance $C = 1 \mu\text{F}$ is suddenly connected to a battery of 100 volt through a resistance $R = 100 \Omega$. The time taken for the capacitor to be charged to get 50 V is:

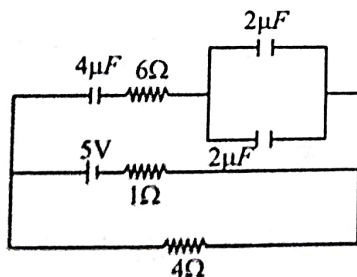
[Take $\ln 2 = 0.69$]

[27 July, 2021 (Shift-I)]



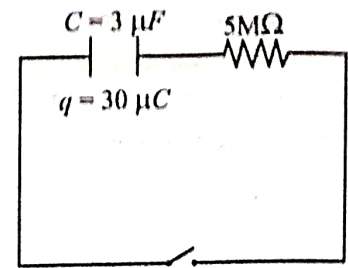
- (a) $3.33 \times 10^{-4} \text{ s}$ (b) $1.44 \times 10^{-4} \text{ s}$
(c) $0.30 \times 10^{-4} \text{ s}$ (d) $0.69 \times 10^{-4} \text{ s}$

226. Calculate the amount of charge on capacitor of $4 \mu\text{F}$. The internal resistance of battery is 1Ω .
[27 Aug, 2021 (Shift-I)]



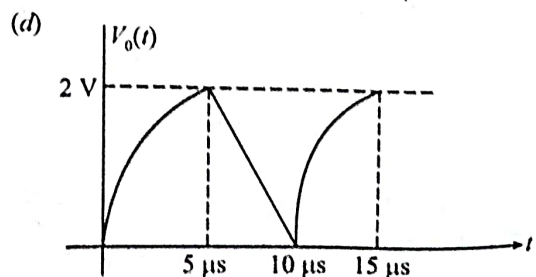
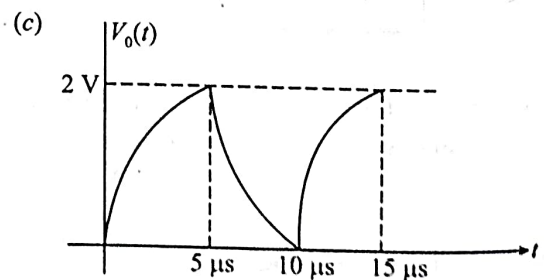
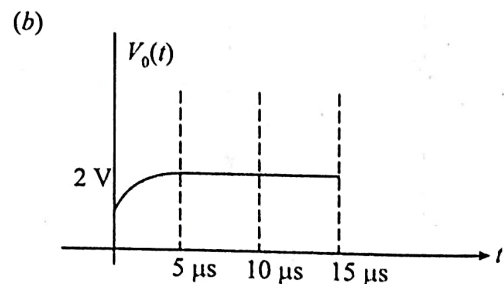
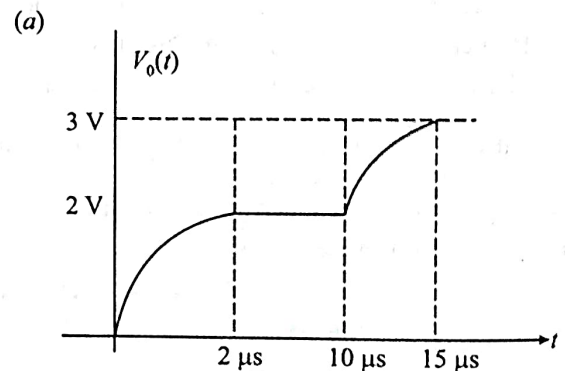
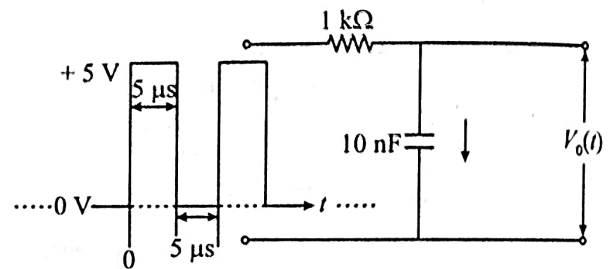
- (a) $16 \mu\text{C}$ (b) zero (c) $4 \mu\text{C}$ (d) $8 \mu\text{C}$

227. The circuit shown in the figure consists of a charged capacitor of capacity $3 \mu\text{F}$ and a charge of $30 \mu\text{C}$. At time $t = 0$, when the key is closed, the value of current flowing through the $5 \text{M}\Omega$ resistor is ' x ' μA .
[18 March, 2021 (Shift-I)]

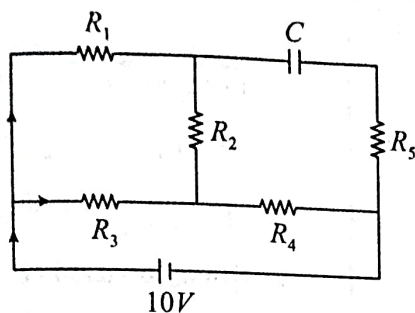


The value of ' x ' to the nearest integer is _____.

228. For the given input voltage waveform $V_{in}(t)$, the output voltage waveform $V_o(t)$, across the capacitor is correctly depicted by
[16 Sep, 2020 (Shift-I)]

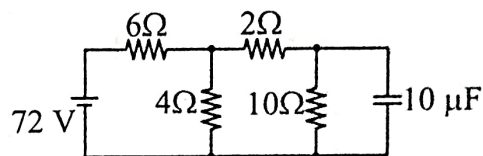


- [2 Sep, 2020 (Shift-11)]



- [7 Jan, 2020 (Shift-II)]

- [9 April, 2019 (Shift-I)]



- (a) $200\ \mu\text{C}$ (b) $60\ \mu\text{C}$ (c) $10\ \mu\text{C}$ (d) $2\ \mu\text{C}$

ANSWER KEY

- [illegible]